Cloud and Virtualization Concepts

Why learn virtualization?

- Modern computing is more efficient due to virtualization
- Virtualization can be used for mobile, personal and cloud computing
- You can also use virtualization in your personal life

THIS CONTENT WILL HELP YOU

- Understand the benefits of virtualization
- Be able to describe virtualization, virtual machines and hypervisors
- Describe typical data center components that are virtualized
- Become familiar with VMware technology popular in industry

VIRTUALIZATION BENEFITS

- Have you ever wished you could clone yourself?
- If you could, would you be more efficient? Would you do more?
- Virtualization enables computers to be more efficient in a similar fashion
- Computers that use virtualization optimize the available compute resources



Simply put, virtualization is an idea that has been around for decades, originating in the 1960s and mainframe computers.

Cloud and Virtualization Concepts will lead you topic-by-topic to build your understanding of:

- **what virtualization is, and how it can be used,**
- **the main components involved in virtualization,**
- **as well as VMware's role in the virtualization industry and,**
- **how knowledge of virtualization can lead to great career opportunities.**

While virtualization has been a part of the IT landscape for decades, it is only recently **(in 1998)** that VMware delivered the benefits of virtualization to industry-standard x86-based platforms,

Course Objectives

- By the end of this course, you should be able to meet the following objectives:
  - Explore creating virtual machines and the benefits
  - Move from a physical environment to a virtual one
  - Understand how individuals and businesses benefit from virtualization
  - Describe virtualization, virtual machines, hypervisors, and VMware vSphere components
  - Describe the concepts of server, network, storage, and desktop virtualization
  - Understand cloud concepts such as types of cloud computing and popular cloud deployment models
  - Understand container technology including concepts such as containers, Docker, and Kubernetes
  - Explore how containers and virtual machines enable cloud computing, DevOps, and cloud-native technology

Course Overview

- The Cloud and Virtualization Concepts course consists of eight modules.
  - **Module 1: Introduction to Virtualization**
  - **Module 2: Introduction to cloud and VMs**
  - **Module 3: The Hypervisor**
  - **Module 4: The Data Center**
  - **Module 5: The Virtual Data Center**
  - **Module 6: The Cloud**
  - **Module 7: Container Technology**
  - **Module 8: VMware Virtualization Solutions**

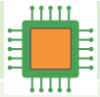
WHAT IS VIRTUALIZATION?

HARDWARE AND SOFTWARE

- Do you use a smartphone, laptop or home computer?
- Smartphones, laptops or home computers are hardware
- Similar to how your brain controls your actions, software controls hardware
- There are different types of software that control computer actions



HARDWARE

Processor 	Also called CPU (Central Processing Unit)
RAM 	Random Access Memory
Read-Only Memory 	Non-volatile memory that stores BIOS *BIOS is a type of software responsible for turning on (booting) computer
Motherboard 	Printed Circuit Board (PCB) that holds the processor, RAM, ROM, network and Input/Output (I/O) and other components
Chipset 	A collection of microchips on motherboard that manage specific functions
Storage 	A persistent (non-volatile) storage device such as a Hard Drive Disk or Solid State Drive

SOFTWARE

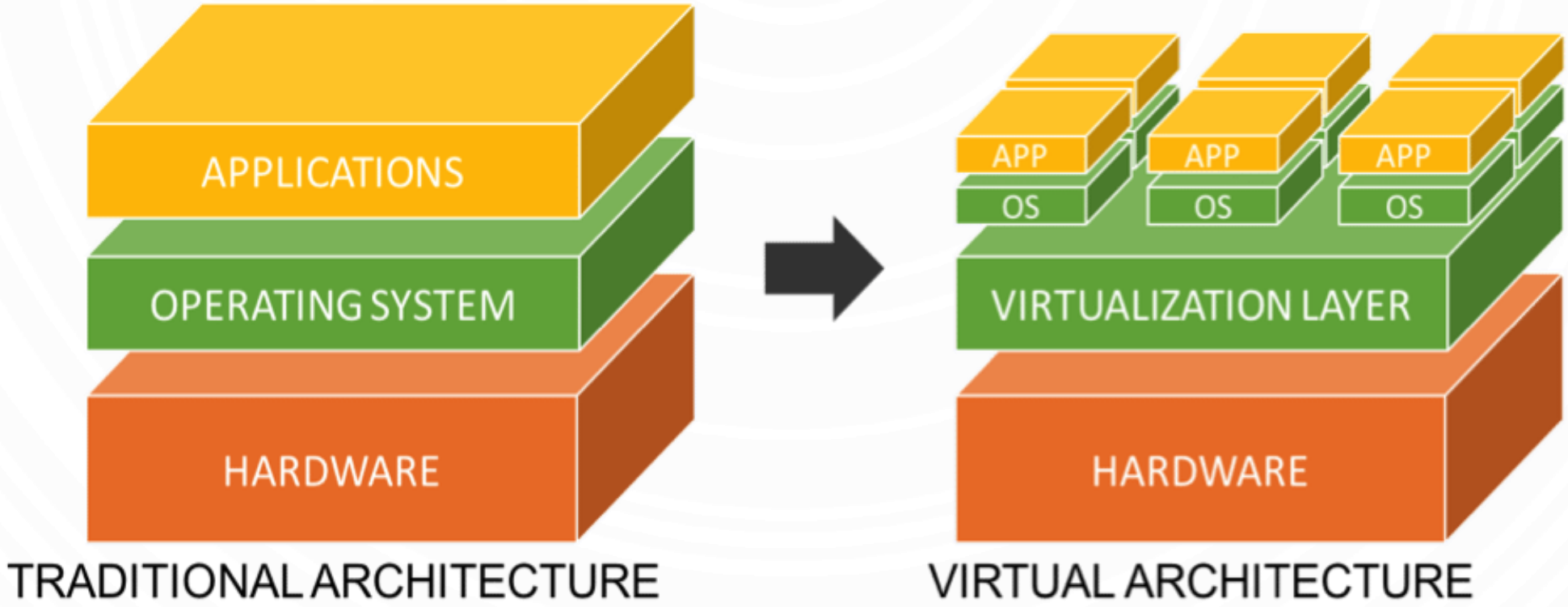
- System software is necessary for hardware to function
- Operating system (OS) controls the hardware



- Application software tells your system to execute a task you want done

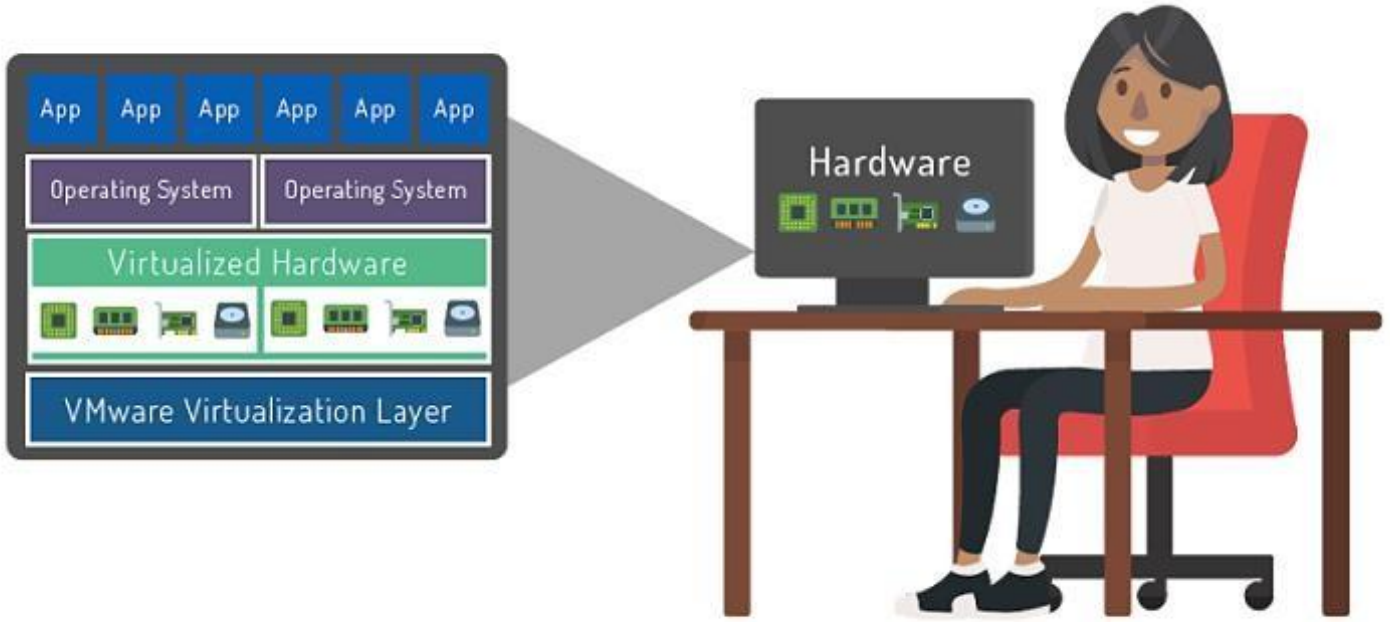
"Virtualization is a technology to run multiple same or different operating systems which is completely isolated from each other."

Ex: Run both Windows and Linux on the same machine



How it is different from dual boot?

Both OSes run simultaneously



Now that you are aware of the roles of hardware and software, the concept of virtualization will be easier to grasp. Virtualization is the "layer" of technology that goes between the physical hardware of a device and the operating system to create one or more copies of the device.

WHAT IS A VM ?

WHAT IS A VM?

- VM stands for “Virtual Machine”
- Virtualization creates virtual hardware by cloning physical hardware
- The **hypervisor** uses virtual hardware to create a virtual machine (VM)

Serial/Parallel Ports

USB Controller

Floppy/Optical Drives and Controllers

Storage Drive







Graphic Card



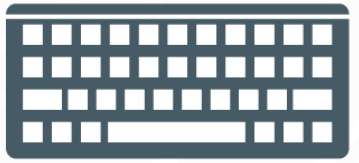
VM



Mouse

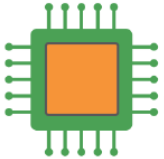
RAM

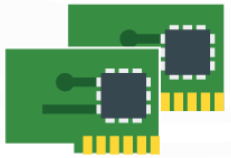




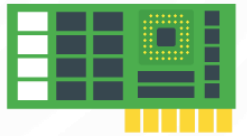
Keyboard

Processor









SCSI/IDE Controllers

NICs

HBAs



- A VM is a set of files
- With a hypervisor running VMs, one computer can run multiple OS simultaneously
- A VM can run applications like a physical machine

CLOUD COMPUTING

- A pool of resources that can be rapidly provisioned in an automated, on-demand manner.
- Value of cloud computing is :
  - Economies of scale
  - Elastic enough to scale with the needs of your organization.
  - Cost and operational benefits
  - Easily accessed by users no matter where they reside

virtualization benefits for our environment

- Using virtual machines results in more compute power from a single server.
- When fewer servers are used, it results in **less energy consumption**.
- The long term outcome of virtualization is less space and power used in data centers.
- Knowing your carbon footprint has environmental and cost benefits.

THE HYPERVISOR

WHAT IS A HYPERVISOR?

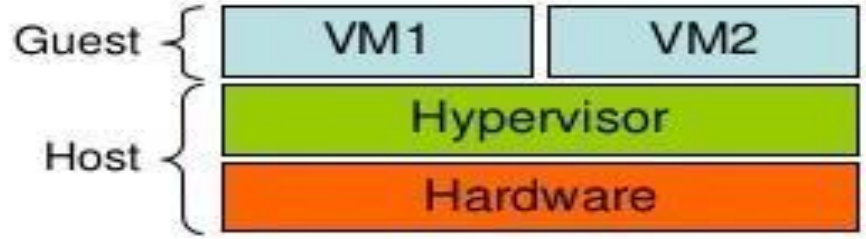
"Hypervisor is a software layer sits between Hardware and OSes which will interact with hardware and resources and provide an interface to share the available resources to Virtual containers."

- Hypervisor is Software installed on top of hardware (a server) that creates the virtualization layer
- Hosts VMs
- The hypervisor pulls physical resources (i.e., CPU, RAM) from the server and turns them into virtual hardware

Types of Hypervisor

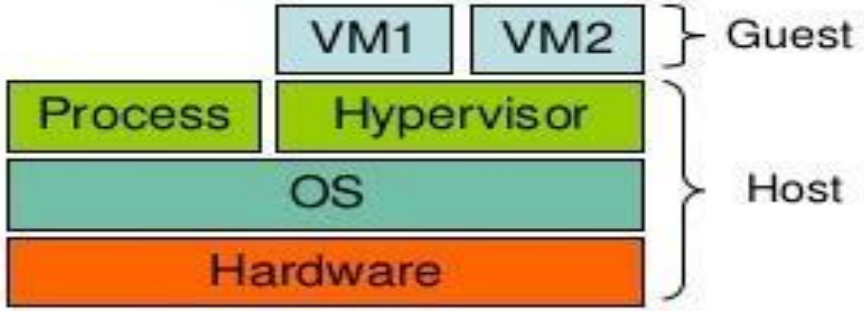
- Bare-Metal
- Hosted

Bare-Metal



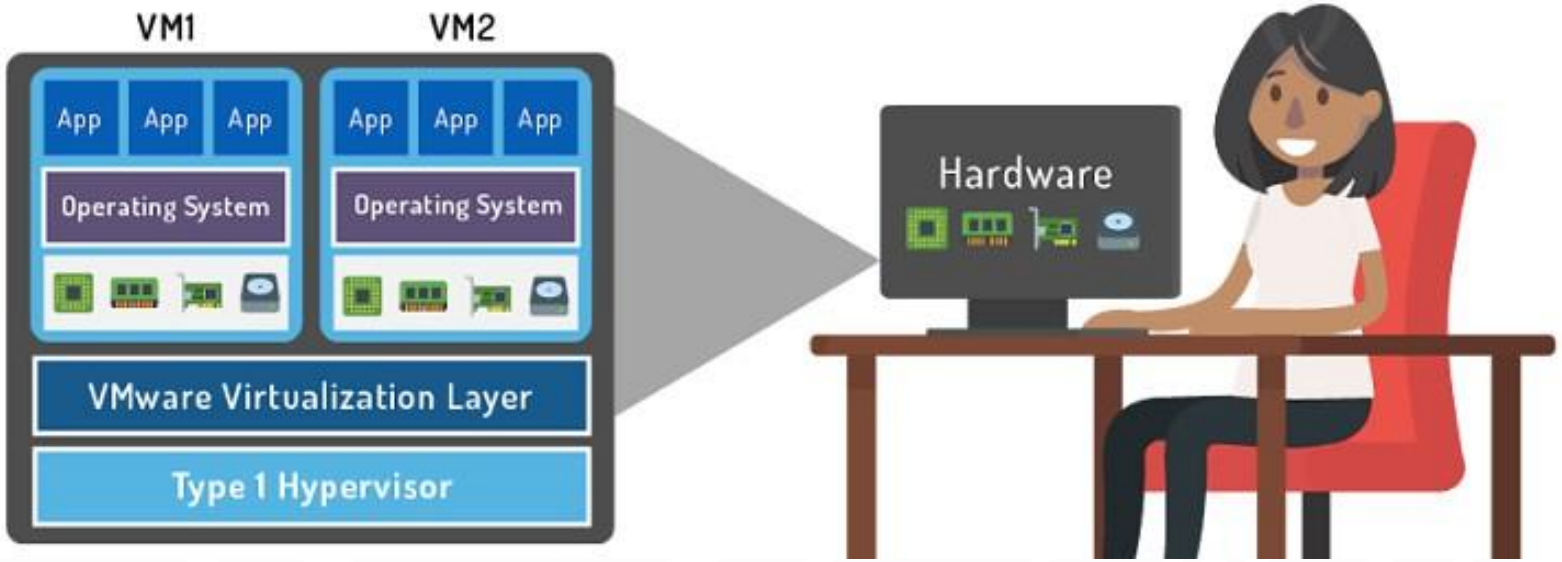
VMware ESX, Microsoft Hyper-V, Citrix XenServer

Hosted

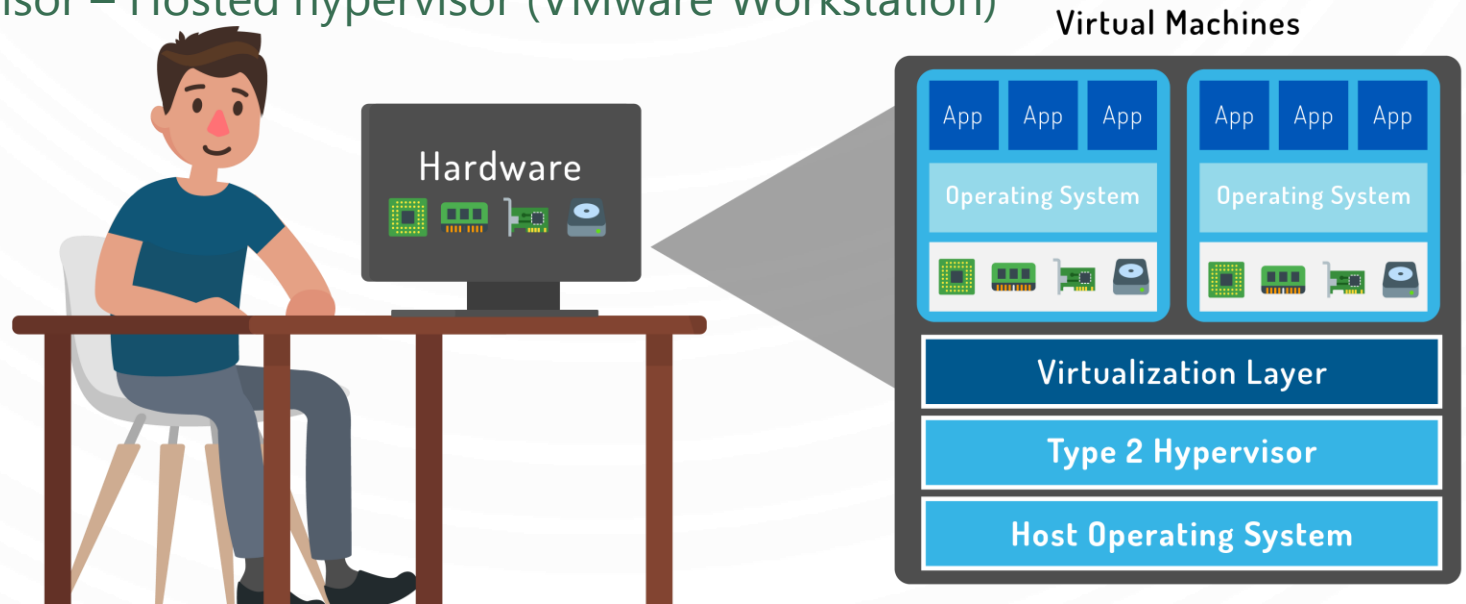


VMware Workstation, Microsoft Virtual PC, Sun VirtualBox, QEMU, KVM

- Type 1 Hypervisor – Bare metal hypervisor (VMware ESXi)



- Type 2 Hypervisor – Hosted hypervisor (VMware Workstation)



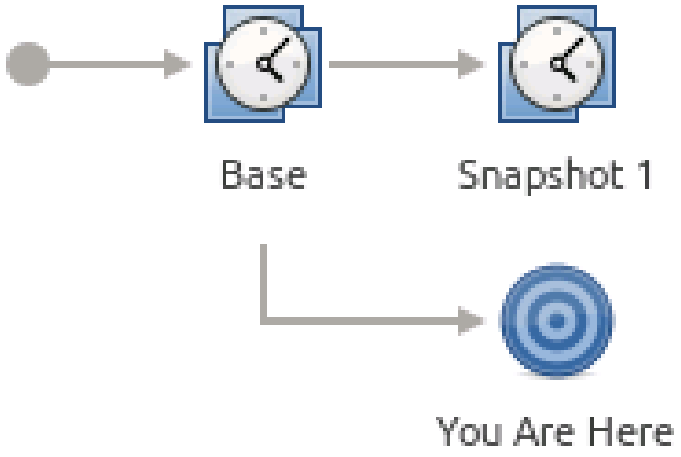
VIRTUAL MACHINE FILES

- VMs can be exported and moved to other hosts
- Files are created by the hypervisor and stored in a directory
- Example for VM files:

File Type	File Name	Description
Log File	<vmname>.log	Keeps a log of VM activity
Disk File	<vmname>.vmdk	Stores content of VM's disk drive
Snapshot Files	<vmname>.vmsd and <vmname>.vmsn	Stores information about VM snapshots (saved VM state)
Configuration File	<vmname>.vmx	Stores information about VM name, BIOS, guest OS, and memory

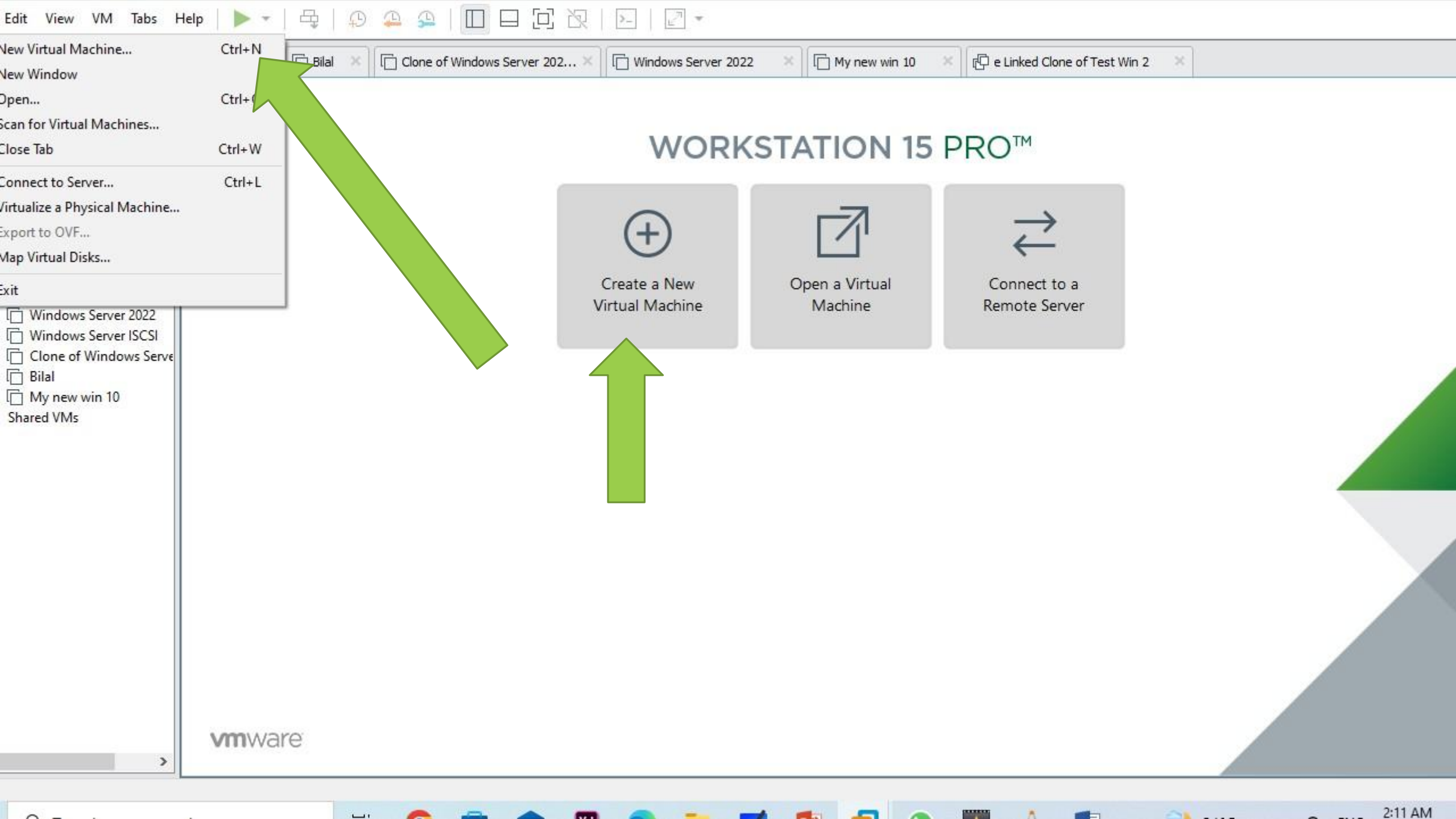
WHAT IS A SNAPSHOT?

- Working on a VM and need to save progress or state?
- Snapshots are saved as files in the VM folder (`<vmname>.vmx`)
- What is saved by a snapshot?
  - State of VM disks
  - Contents of VM memory
  - VM settings



INSTALL VMWARE

CREATE VIRTUAL MACHINE ON VMWARE


Edit View VM Tabs Help
New Virtual Machine... Ctrl+N
New Window
Open... Ctrl+O
Scan for Virtual Machines...
Close Tab Ctrl+W
Connect to Server... Ctrl+L
Virtualize a Physical Machine...
Export to OVF...
Map Virtual Disks...
Exit
Bilal
Clone of Windows Server 202...
Windows Server 2022
My new win 10
e Linked Clone of Test Win 2
WORKSTATION 15 PRO™
Create a New Virtual Machine
Open a Virtual Machine
Connect to a Remote Server
Windows Server 2022
Windows Server ISCSI
Clone of Windows Serve
Bilal
My new win 10
Shared VMs
vmware
2:11 AM

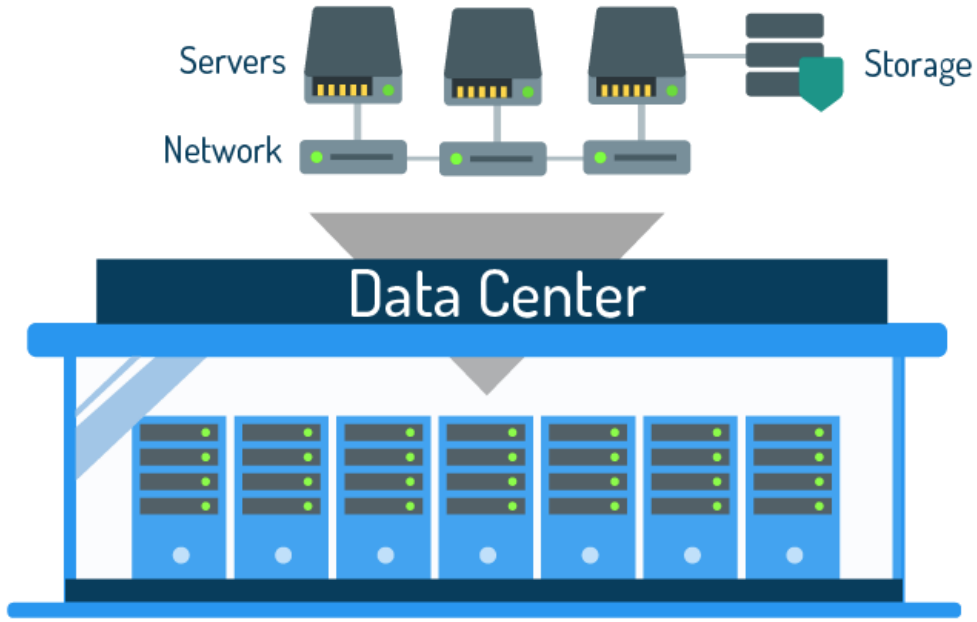
BOOTING FROM VMWARE

INSTALL WINDOWS 10 ON VMWARE

THE DATA CENTER

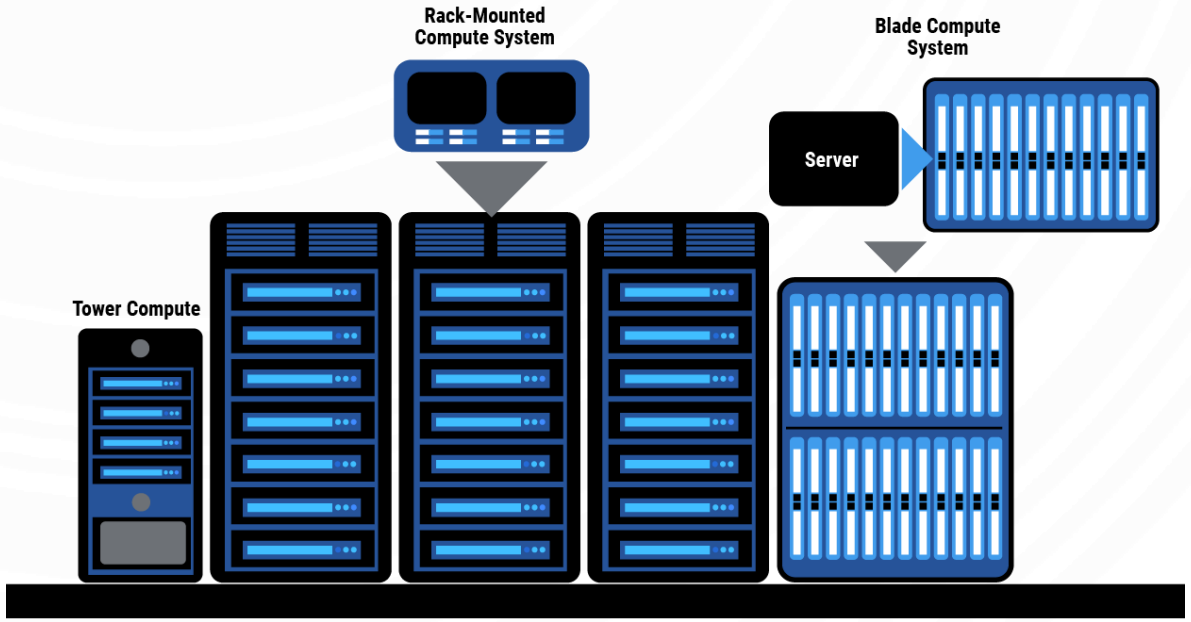
WHAT IS A DATA CENTER?

- **Hardware** infrastructure that supports virtualization
- Focus is on processing large amounts of data
- What are the three main components?
  - Compute
  - Storage
  - Networks



COMPUTE SYSTEMS

- Hardware and operating system software that runs applications
- Difference between a PC and a server
  - PCs have a user-friendly interface while servers focus on running programs
- Types of servers:
  - Tower
  - Blade server
  - Rack-mounted server

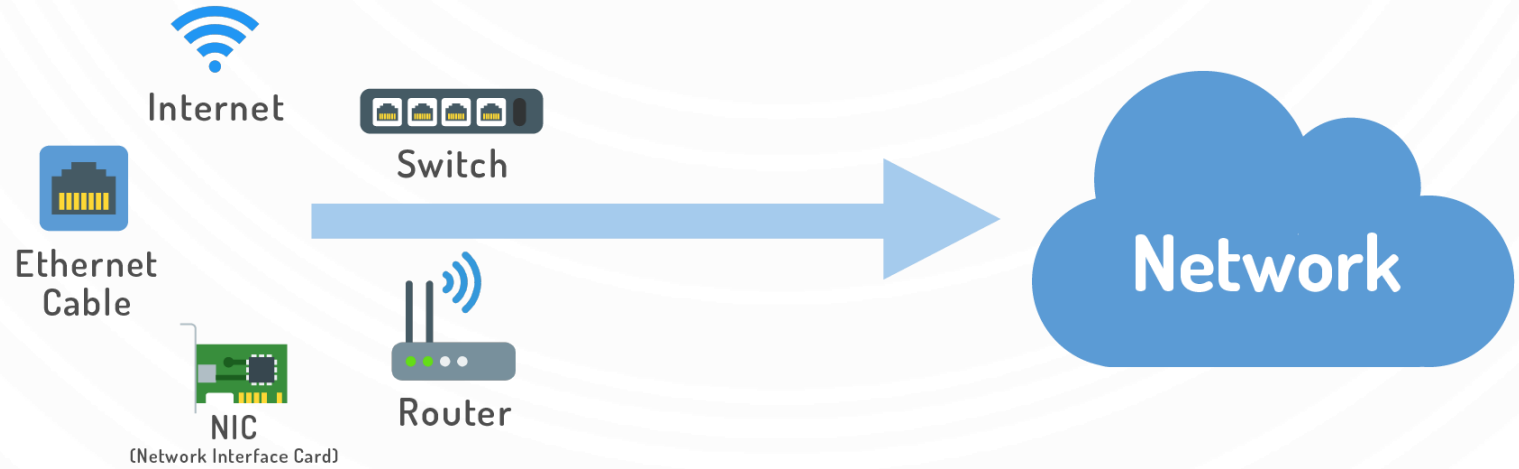


COMPUTE SYSTEMS

- What is the architecture of a server?
- Architecture means the type of processor used by the server.
- Data centers typically use servers with an x86 architecture.
- Classic architecture is 32-bit and in 2003 the 64-bit extension was released for x86 architecture.
- Virtualization technology is compatible with 64-bit but not 32-bit.

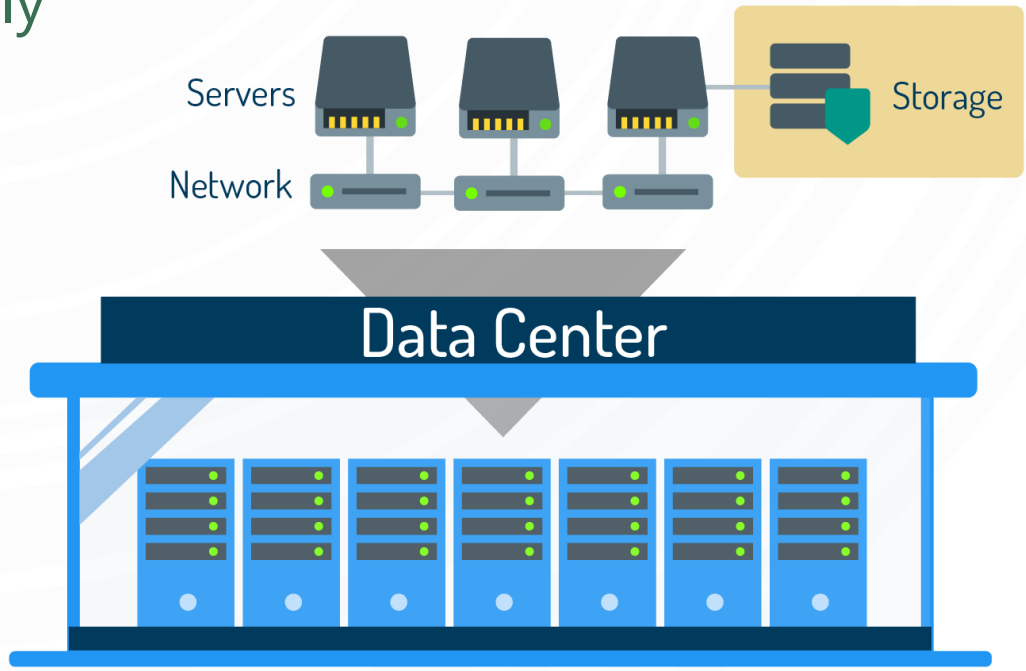
NETWORKS

- Transfer data across the data center so devices can communicate
- What type of hardware is used for networking?
  - Switches, Network Interface Card (NIC), Router, Ethernet Cable



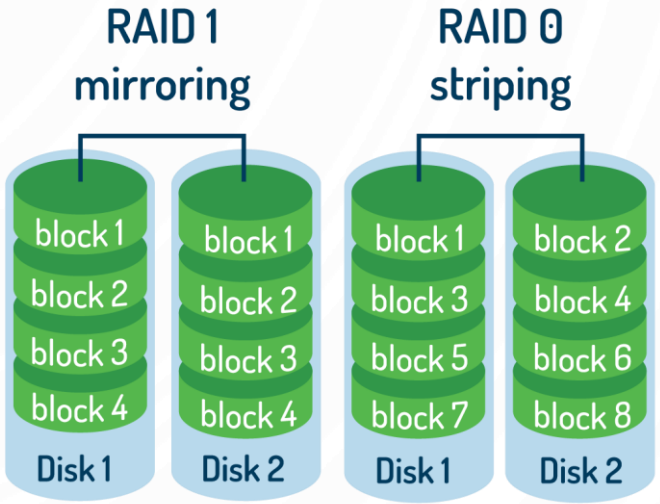
STORAGE

- A data center needs to accommodate high levels of data
- Compute memory is not adequate
- Data centers use devices built specifically for storing data
- Data center storage should have two features: **availability** and **redundancy**



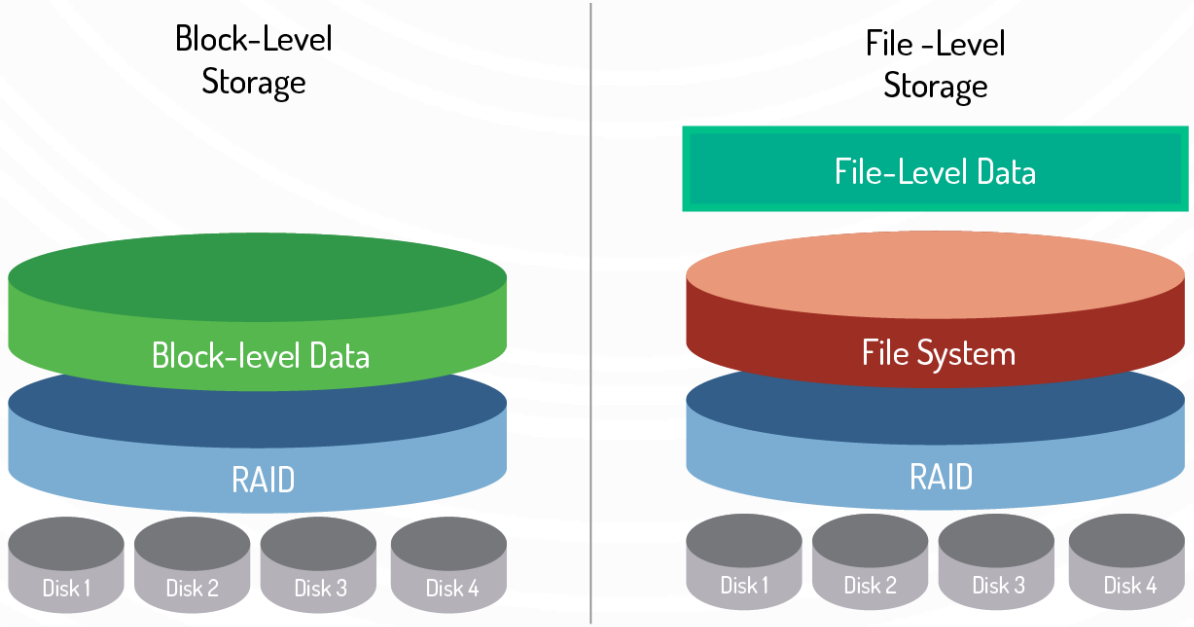
STORAGE - RAID

- **R**edundant **A**rray of **I**ndependent **D**isks
- Hard drives linked together to create a large volume of **redundant** storage
- What are the three methods of writing to RAID?
  - Mirroring
  - Striping
  - Parity
- What do the RAID numbers mean (i.e., 0, 1, 5)?
- Levels of RAID are used to identify types of RAID and what storage availability and redundancy they offer (i.e., 1 = mirroring, 5 = parity)



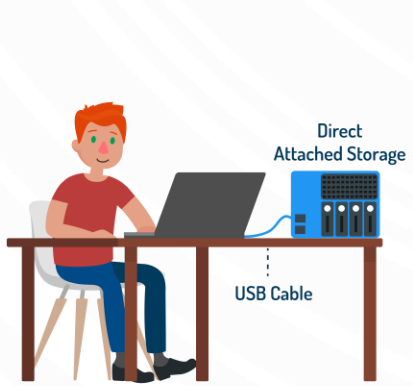
STORAGE - BLOCK VS. FILE-LEVEL

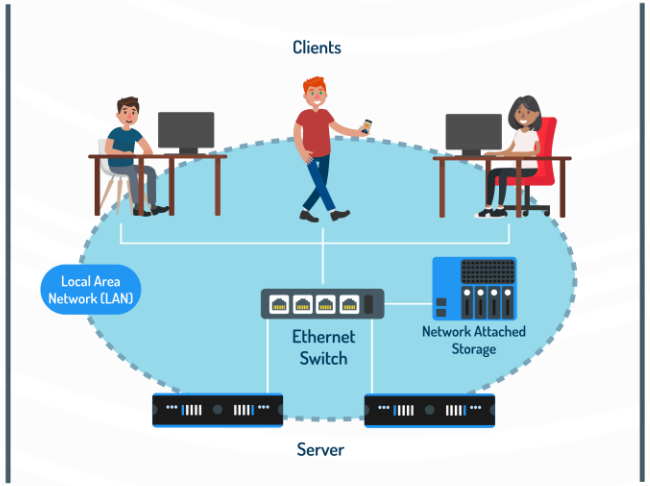
- **Block-Level Storage** – Data is written to and accessed from storage volumes (blocks)
- **File-Level Storage** – Data is written to disks but accessed from the default file system

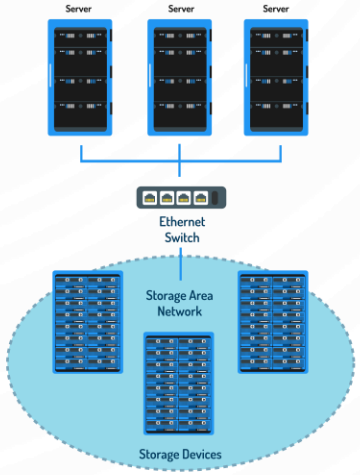


STORAGE – TYPES OF DATA CENTER STORAGE

- **DAS (Direct Attached Storage)** – Storage device is directly attached to a server (block-level)
- **NAS (Network Attached Storage)** – Storage device is attached to a network, servers on the network can access device (file-level)
- **SAN (Storage Area Network)** – Clustered storage devices on their own network that servers can connect to (block-level)





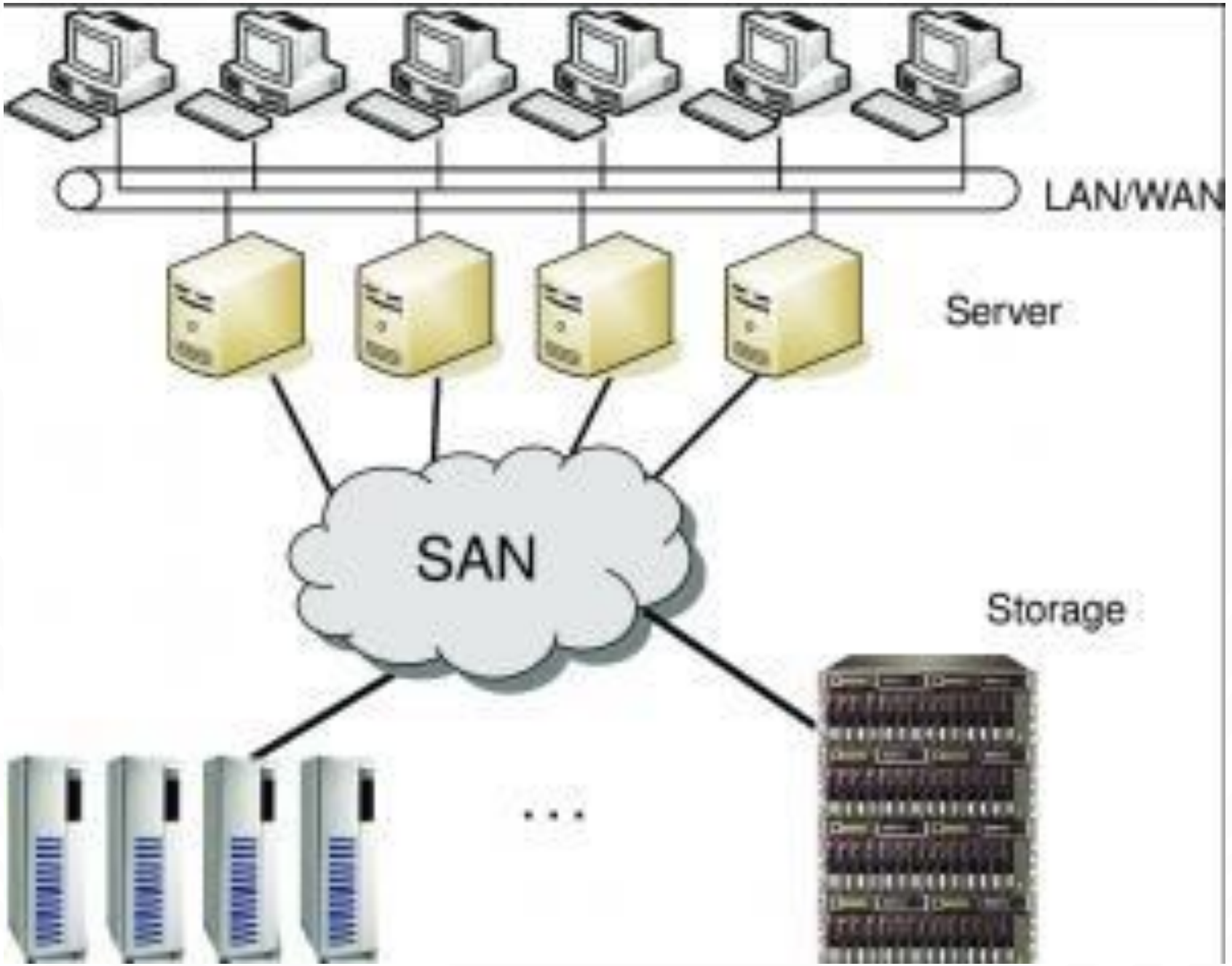


STORAGE TYPE

NAS (Network Attached Storage)



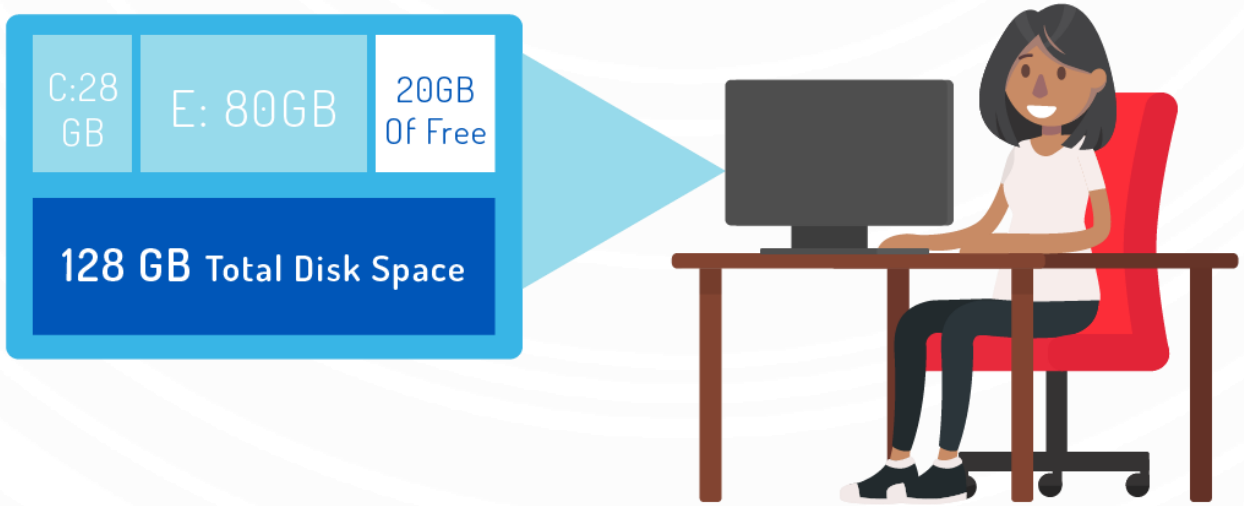



LAN/WAN
Server
SAN
Storage
...

Protocol	Application
SCSI (Small Computer System Interface)	Medium- sized blade servers, Enterprise servers, DAS
SATA (Serial Advanced Technology Attachment)	Small-sized tower server, DAS
eSATA (external SATA)	Small-sized tower server, DAS
SAS (Serial Attached SCSI)	Medium-sized blade servers, Enterprise servers, DAS
FC (Fiber Channel) (single-mode) (SMF)	Enterprise servers, SAN
FC (Fiber Channel) (multi-mode) (MMF)	Enterprise servers, SAN
FCoE (Fiber Channel over Ethernet)	Enterprise servers, SAN
iSCSI (internet Small Computer System Interface)	Enterprise servers, NAS

STORAGE PROVISIONING

- **Thick provisioning:** Disk space is strategically pre-allocated to a server, or a VM.
- This means that the logical space provided by partitioning is equal to the amount of actual physical space set aside on the physical disk.



Thick Virtual Disk
(Full provision all the time)

Thin Virtual Disk
(Over-subscribed provisioning, dynamic size)



VMFS Volume/Database
provisioned for ESX

THE VIRTUAL DATA CENTER

BENEFITS OF A VIRTUAL DATA CENTER

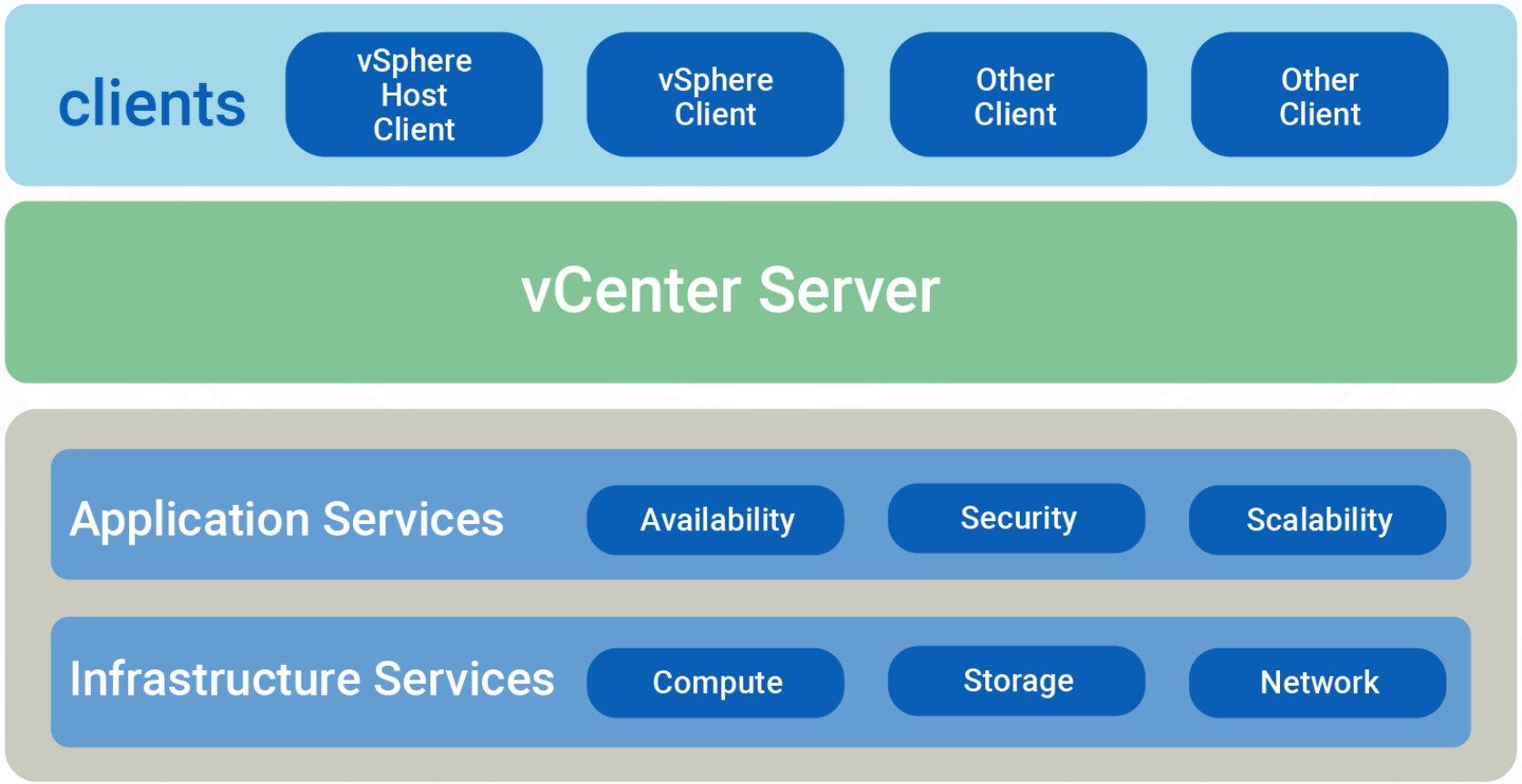
- Data centers use a lot of hardware and virtualization makes hardware more efficient
- Increased computing resources results in higher **availability** of applications
- Less labor needed to monitor data center (administrator can monitor from their desk using a program)
- **Software-defined data center (SDDC):** Hypervisor pools physical data center resources into a virtual data center

WHAT IS VSPHERE?

- Suite of virtualization technology designed for larger enterprise data center management

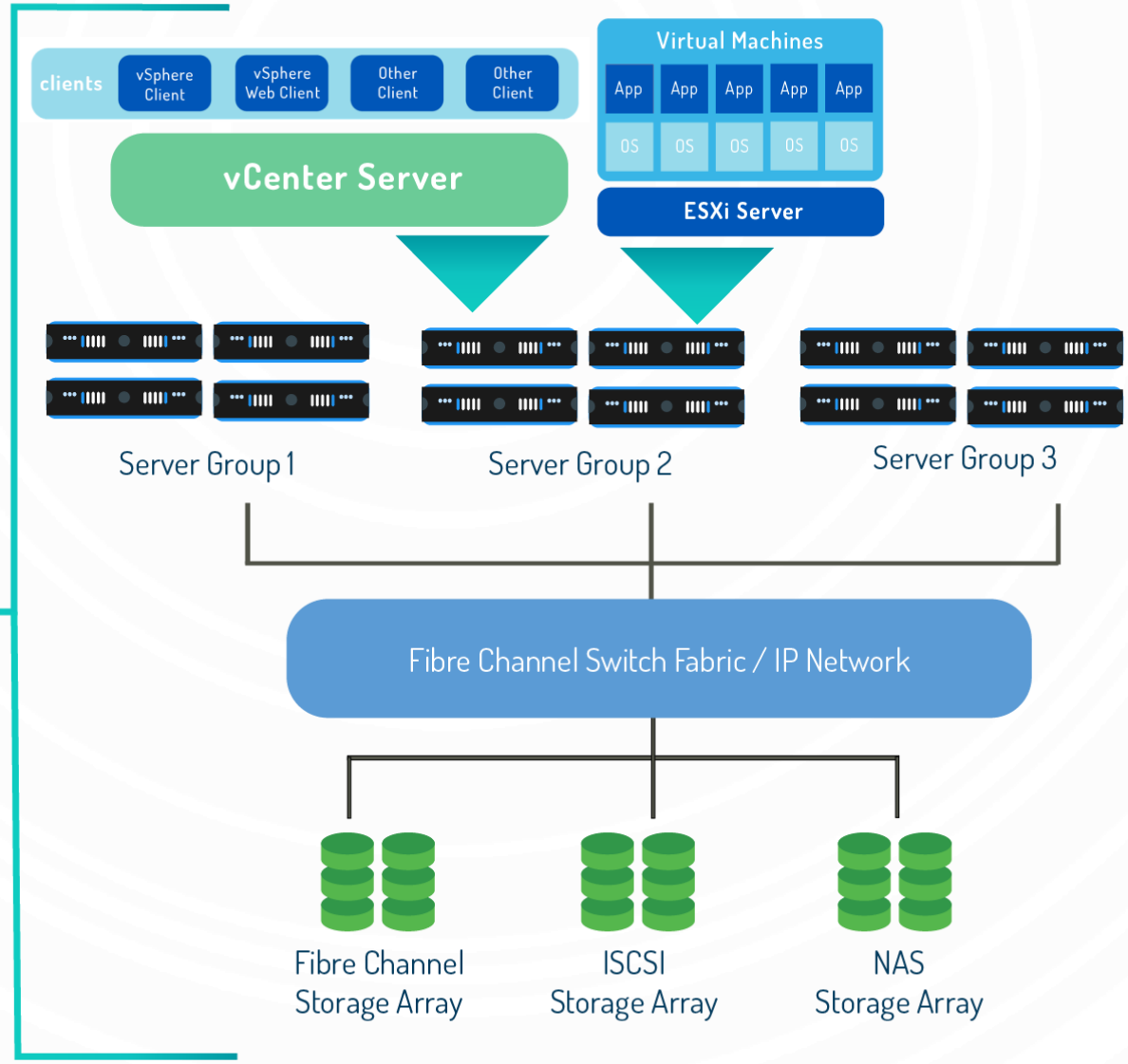
- vSphere virtualization tools include:
  - **ESXi**: Type 1 Hypervisor
  - **vCenter**: Management software (installed on management server)
  - **vSphere Client**: Program that configures the vCenter, host, and operates its virtual machines
  - **vSphere Host Client:** Program that only configures the host and operates its virtual machines

WHAT IS VSPHERE?



VSPHERE

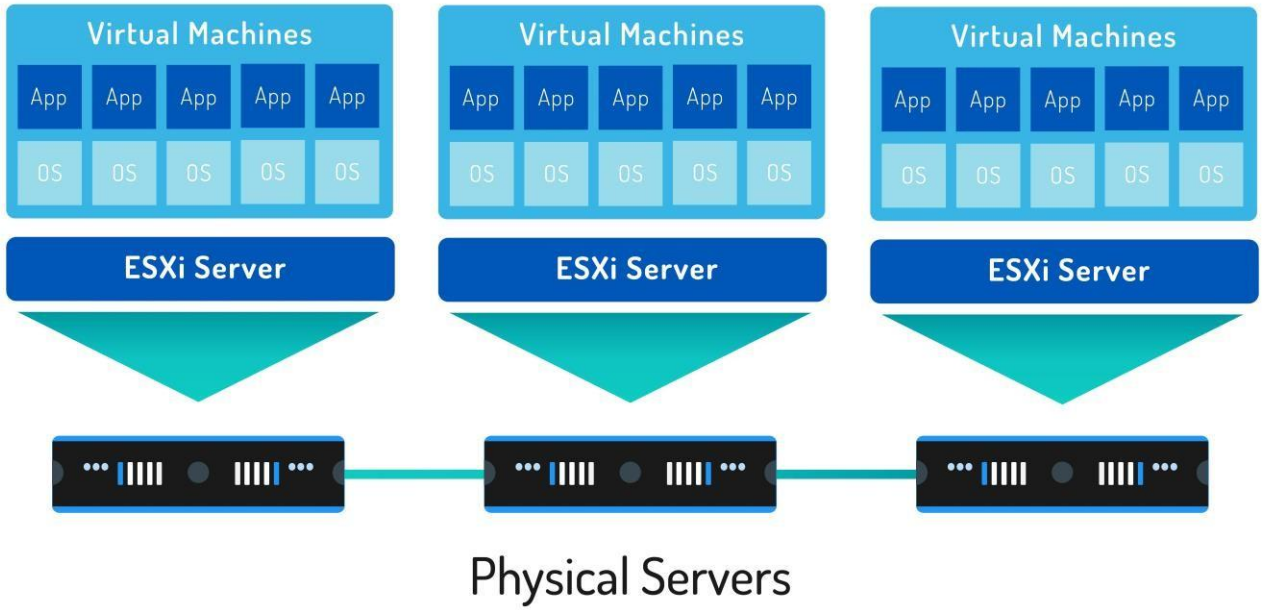

Data Center


clients
vSphere Client
vSphere Web Client
Other Client
Other Client
Virtual Machines
App
App
App
App
App
OS
OS
OS
OS
OS
vCenter Server
ESXi Server
Server Group 1
Server Group 2
Server Group 3
Fibre Channel Switch Fabric / IP Network
Fibre Channel Storage Array
ISCSI Storage Array
NAS Storage Array

WHAT IS ESXI?

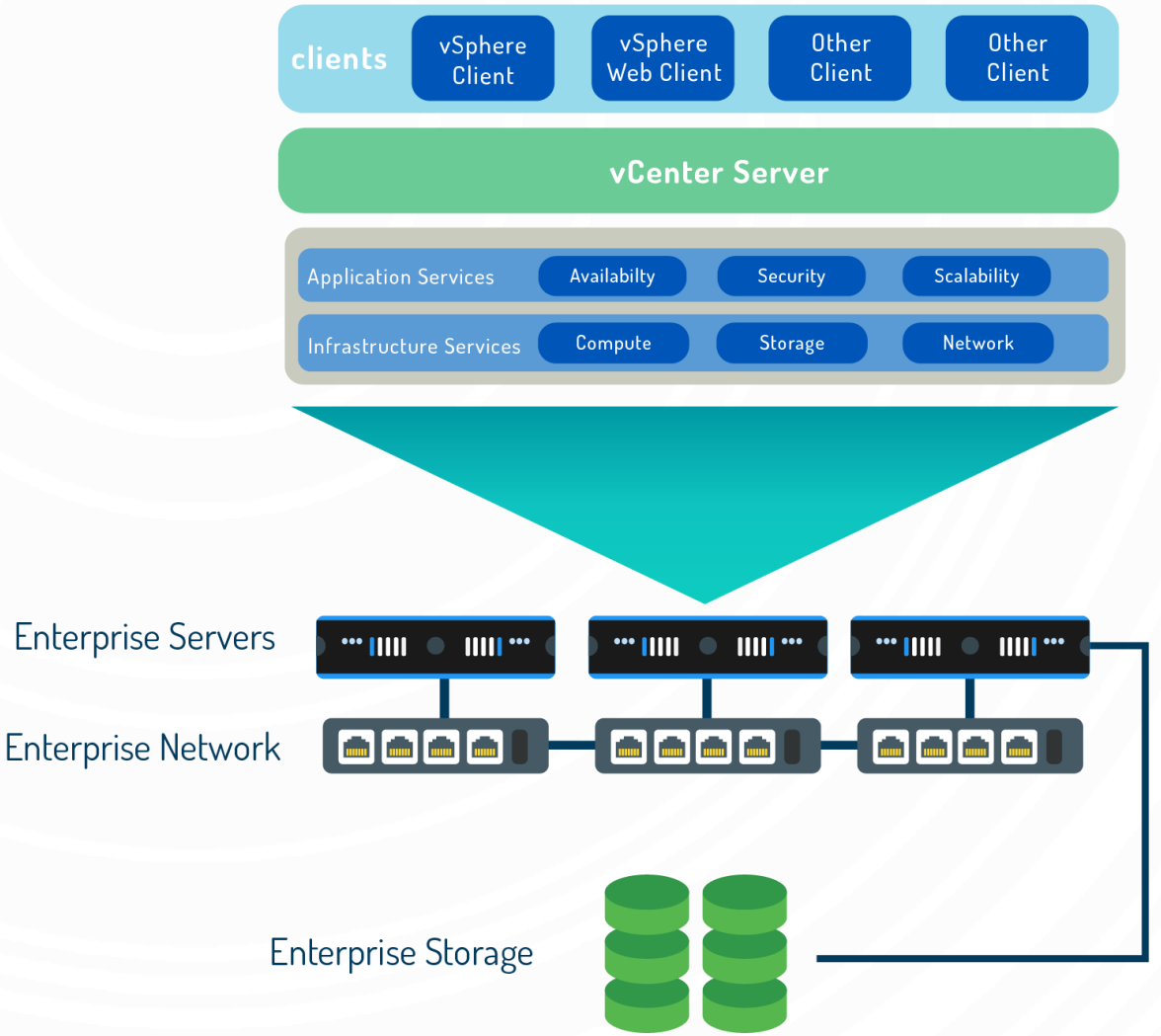
- ESXi is VMware's Type 1 hypervisor software installed directly on the physical server and creates the virtual layer

- Components of ESXi:
  - Unix Microkernel
  - VMware Kernel (VMkernel)



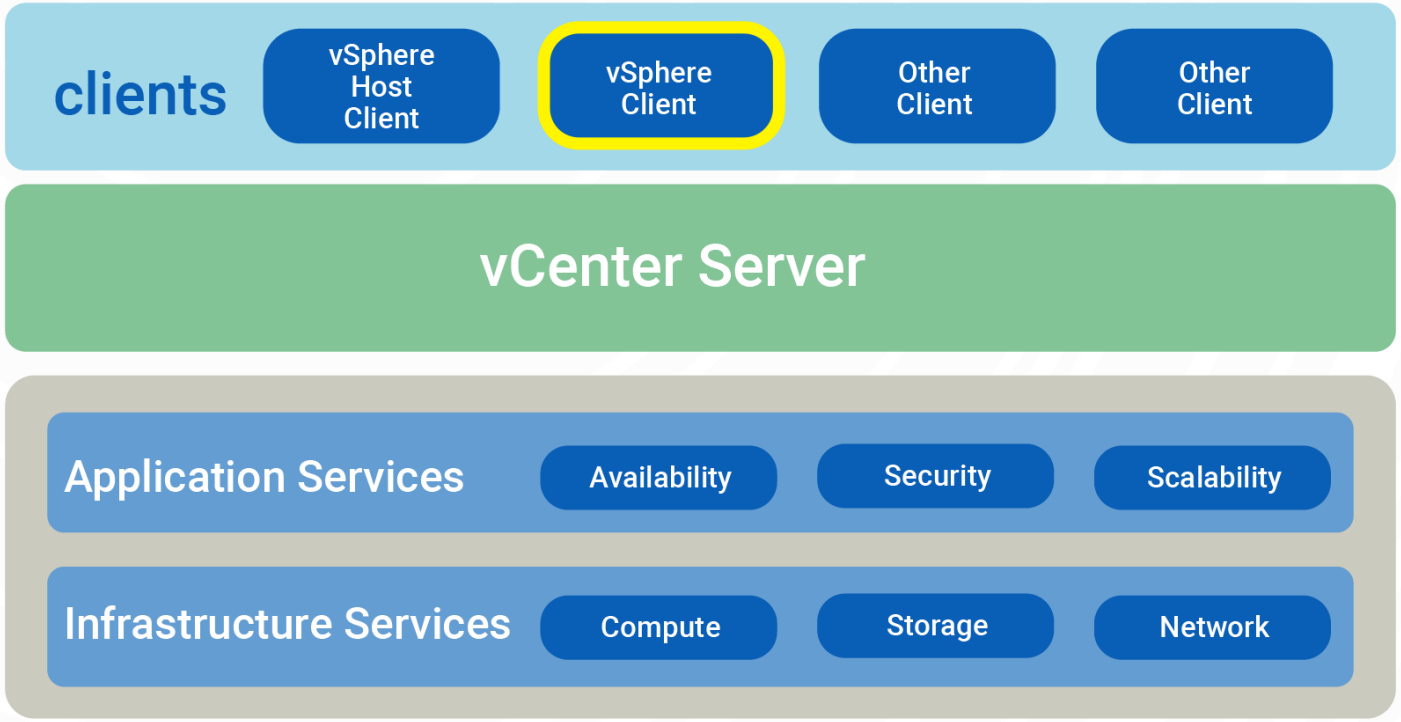
VCENTER

- **vCenter**: Software installed on a dedicated server to manage ESXi servers and other components of a virtualized data center.

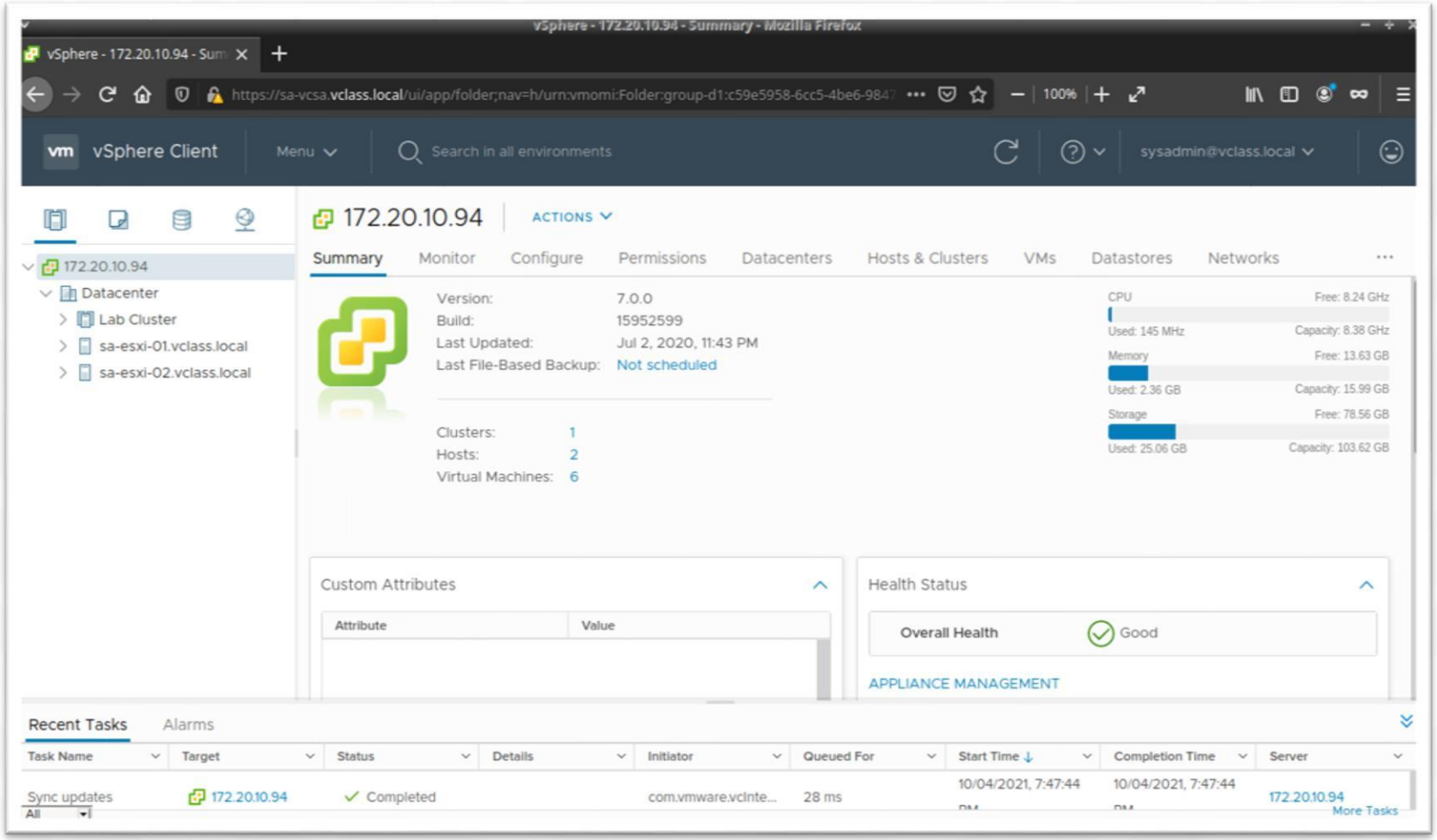


VSPHERE CLIENT

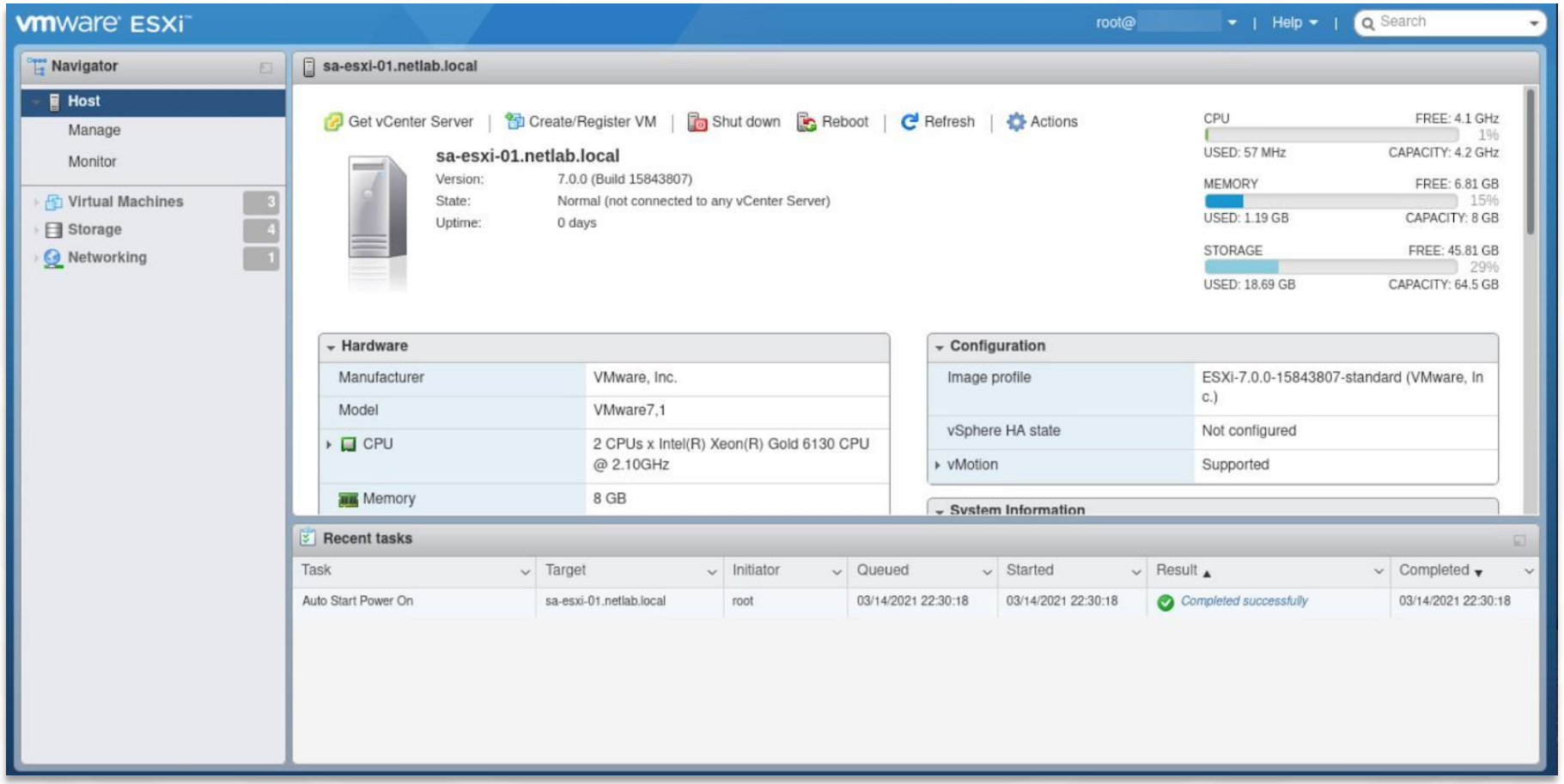
- **vSphere Client**: Program with a graphical user interface (GUI) that allows data center administrators to connect to vCenter and ESXi remotely.



VSPHERE CLIENT

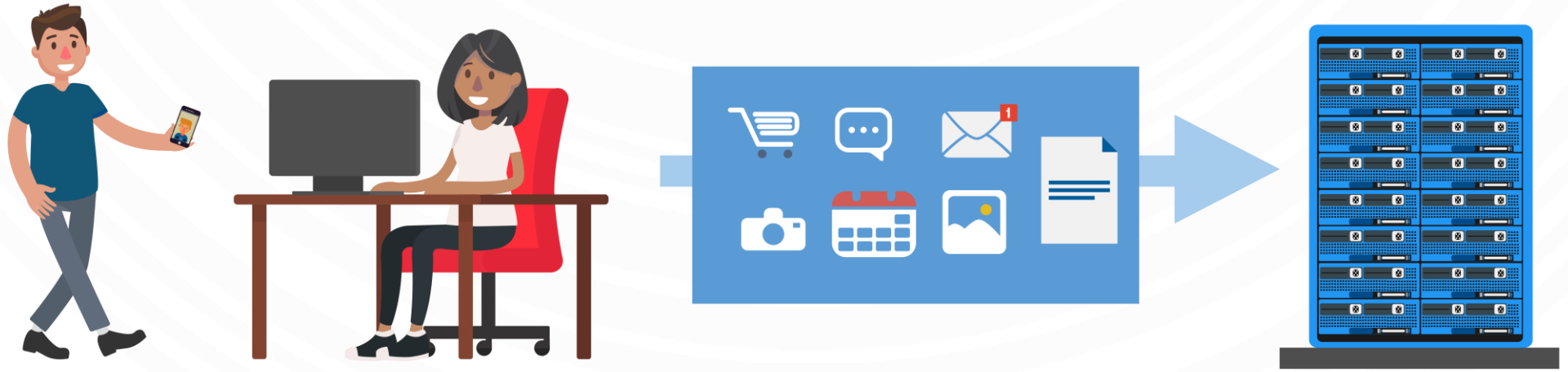


VSPHERE HOST CLIENT



SERVER VIRTUALIZATION

- Results in increased efficiency of data center servers because multiple VMs can be hosted on one server
- Computing resources can be distributed to customers using less hardware

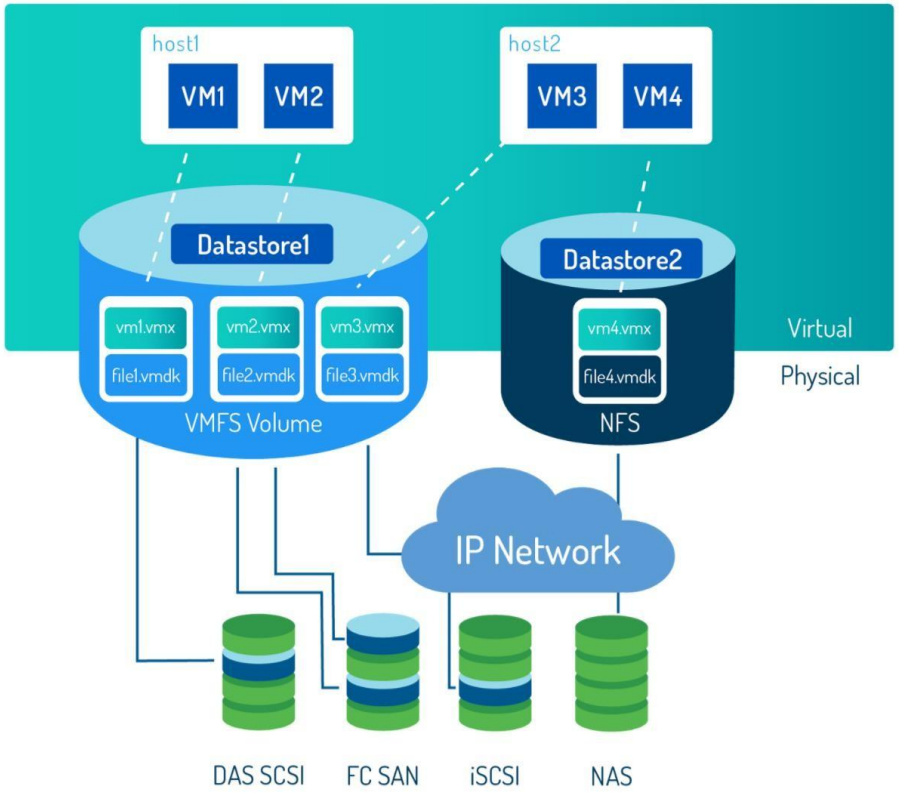


STORAGE VIRTUALIZATION

STORAGE VIRTUALIZATION

- Storage capacity is pooled and distributed to the VMs
  - Physical storage devices are partitioned into logical storage (**LUNs**)
  - LUNs are used to create a **datastore**

- How do VMs access data center storage?
  - VMs are stored as VMDK (*.vmdk*) files on datastore
  - VM configuration files (VM settings) are stored as VMX (*.vmx*) files

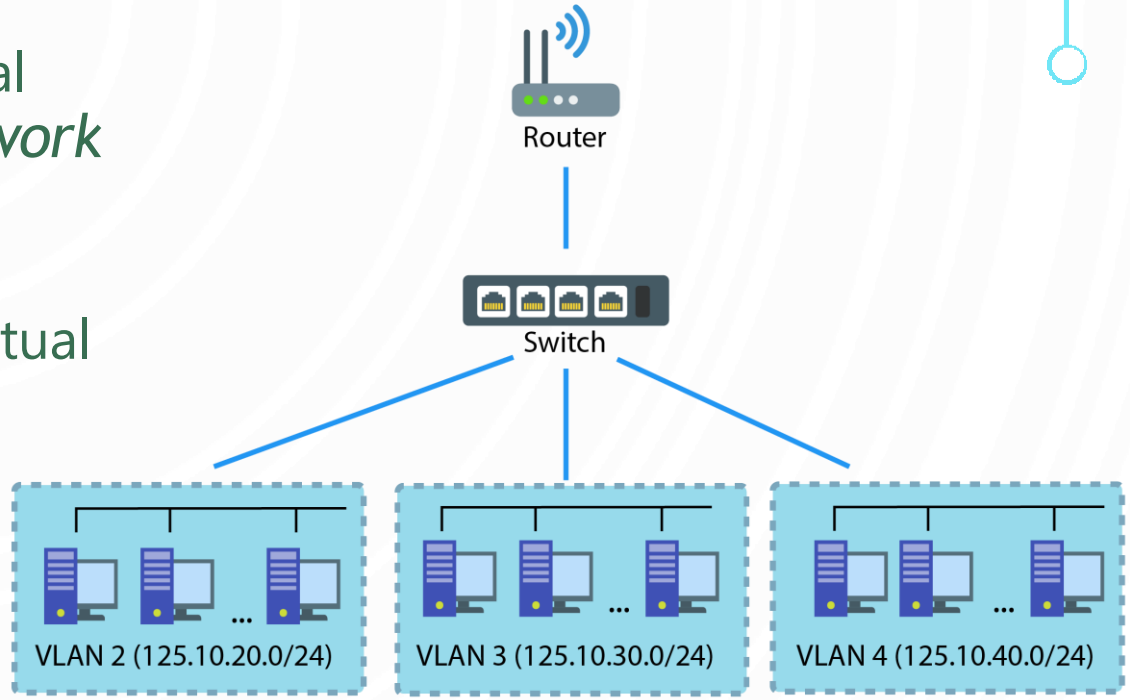


NETWORK VIRTUALIZATION

- Physical components that make up the physical network are virtualized to create a *virtual network*

- What is a vSwitch?

  - Virtual switch that virtual devices can connect to in order to communicate with each other

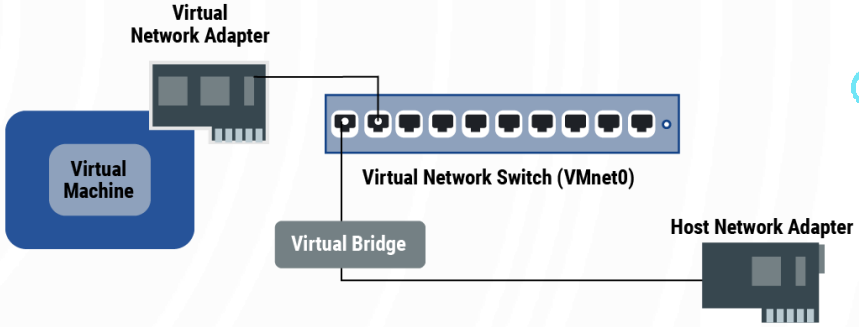


- What is a vLAN?

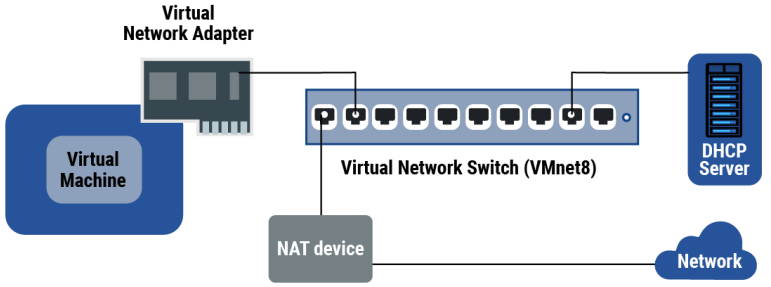
  - Virtual Local Area Network that is segmented into groups of ports isolated from one another, creating different network *segments*

TYPES OF VIRTUAL NETWORKS

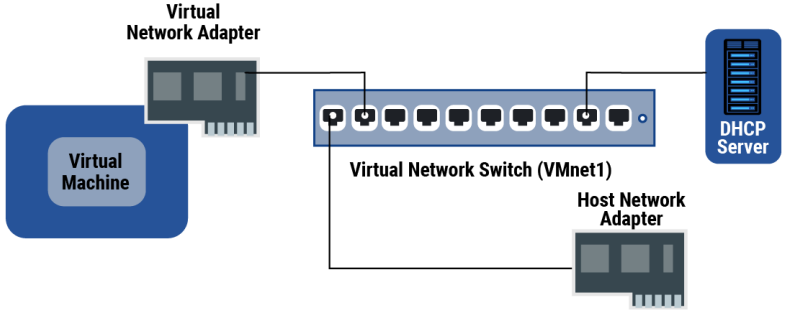
- **Bridged Network**: The host server and the VM are connected to the same network, and the host shares its IP address with the VM



- **NAT**: VMs use an IP translated from the host's IP (using NAT device) and communicate on a private network set up on the host computer



- **Host-only Network**: VMs use a private network but do not have translated IP addresses to connect to an external network, therefore can only communicate to other VMs on the isolated host network



APPLICATION AND DESKTOP VIRTUALIZATION

- Why use virtualized applications?
  - Some applications have specific system requirements
  - VMware **ThinApp** creates a packaged virtual app, that contains the program and system requirements, and delivers it to the end-user

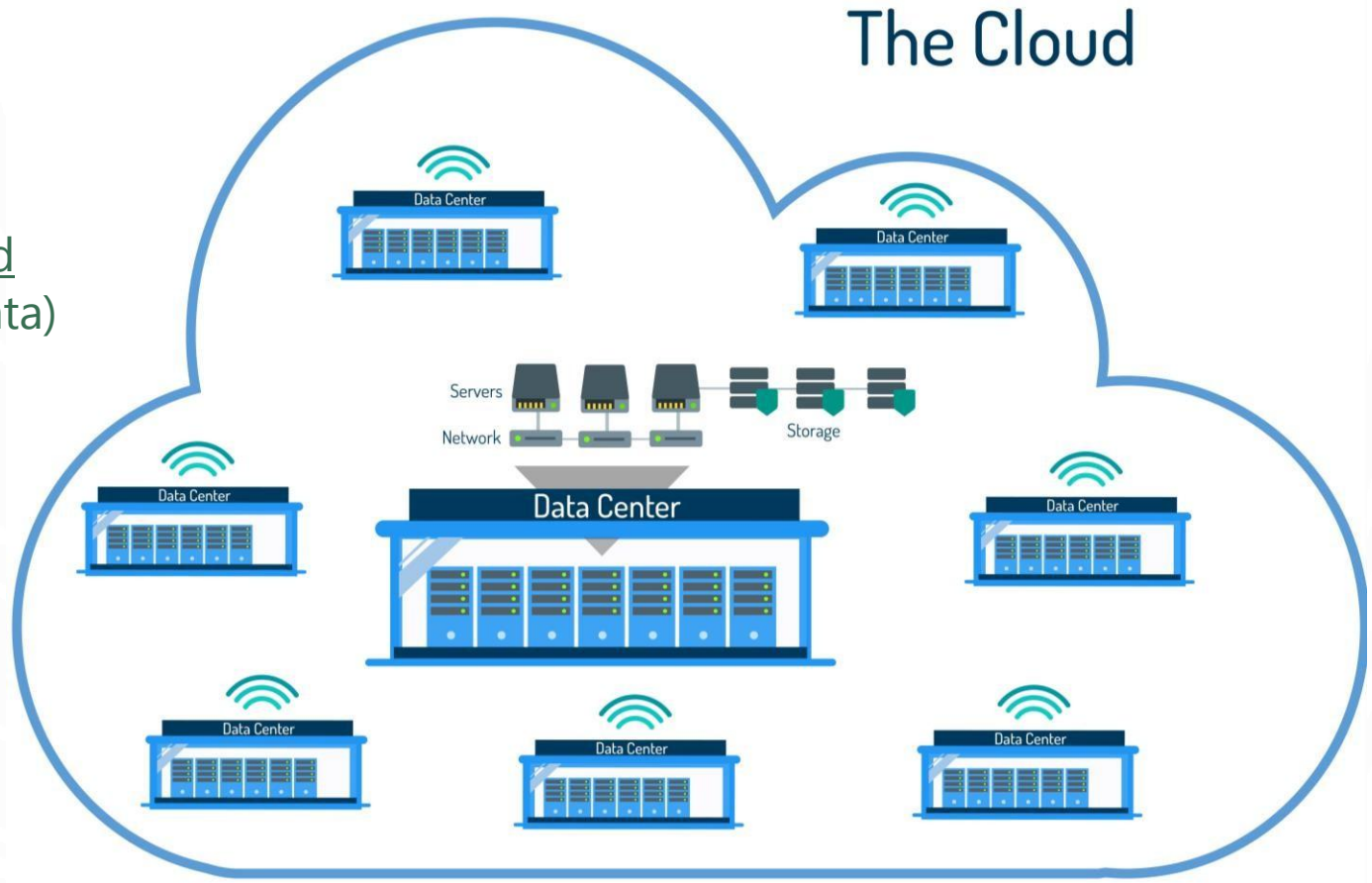
- What is desktop virtualization?
  - Designed to solve computing resource issues faced by the mobile workforce (workers that need computing without the hardware)
  - VMware **Horizon** takes the resources needed to create a desktop environment from data centers and delivers it to the end-user's device

WHAT IS CLOUD ???

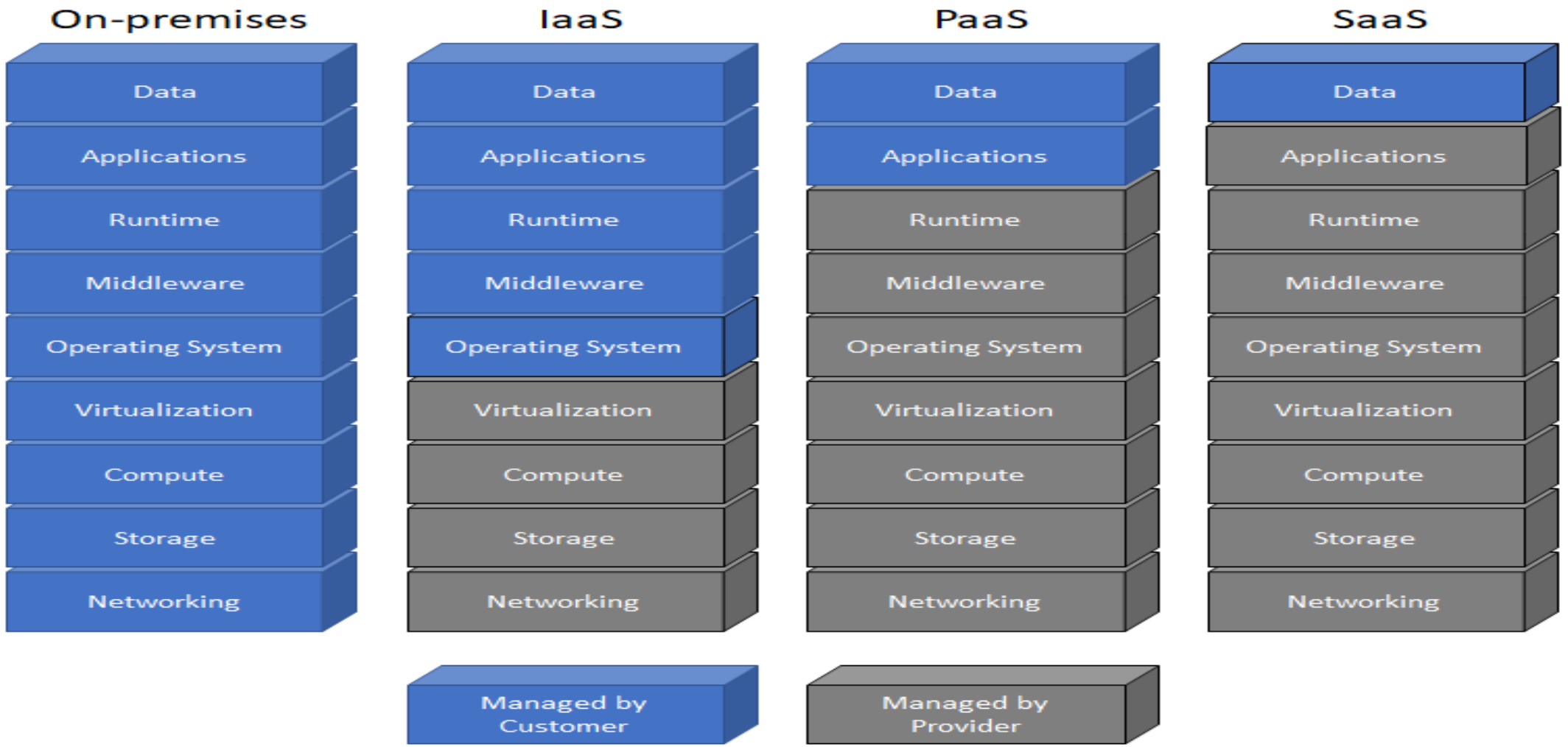


THE CLOUD

- Cloud computing is the delivery of shared computing resources (software and/or data) on-demand through the internet



Cloud computing service models

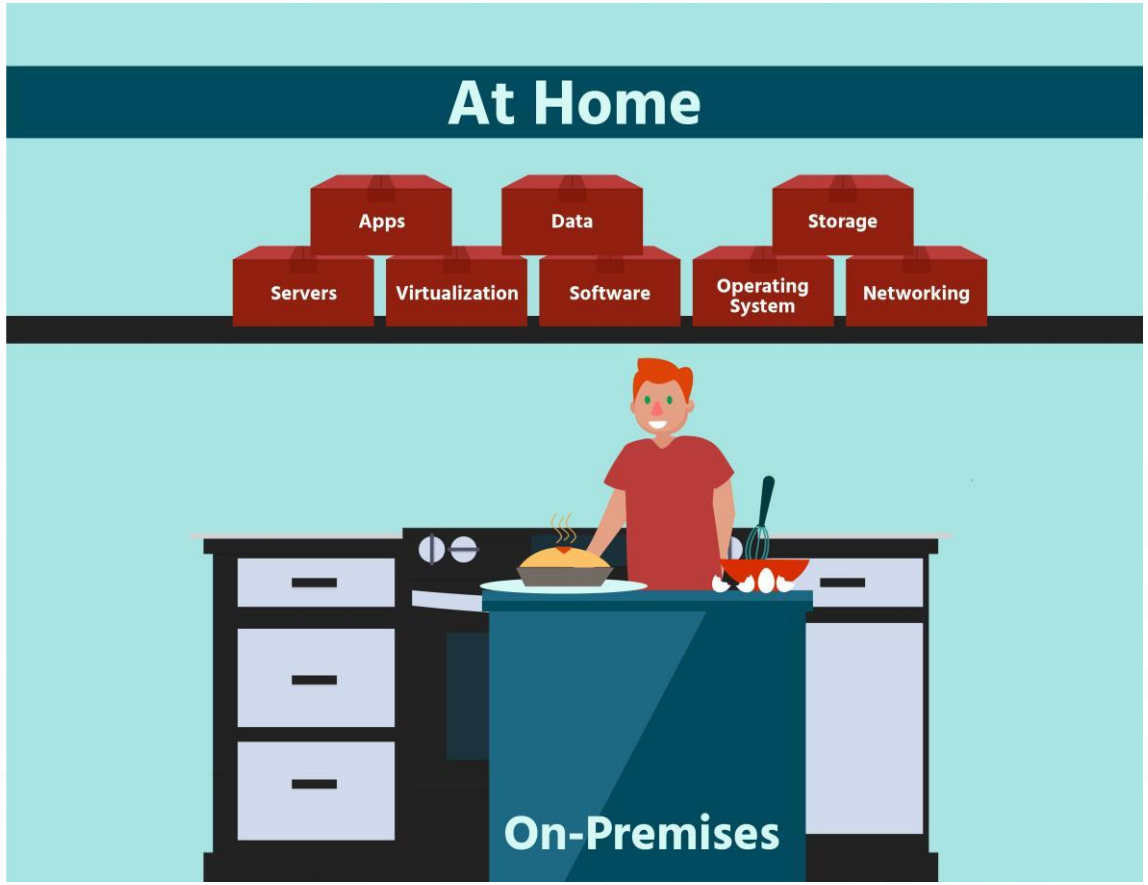


CLOUD COMPUTING SERVICE MODELS

- **Software as a service (SaaS).**
  - Customers are provided access to **an application** running on a cloud infrastructure.
  - but the customer has no knowledge of, and does not manage or control, the underlying cloud infrastructure.
- **Platform as a service (PaaS).**
  - Customers can **deploy supported applications** onto the provider's cloud infrastructure,
  - but the customer has no knowledge of, and does not manage or control, the underlying cloud infrastructure.
  - The company owns the deployed applications and data, and it is therefore responsible for the security of those applications and data.
- **Infrastructure as a service (IaaS).**
  - Customers can provision **processing, storage, networks, and other computing resources**, and **deploy and run operating systems and applications.**
  - the customer has no knowledge of, and does not manage or control, the underlying cloud infrastructure. The customer has control over operating systems, storage, and deployed applications, along with some networking components (for example, host firewalls).
  - The company owns the deployed applications and data, and it is therefore responsible for the security of those applications and data.

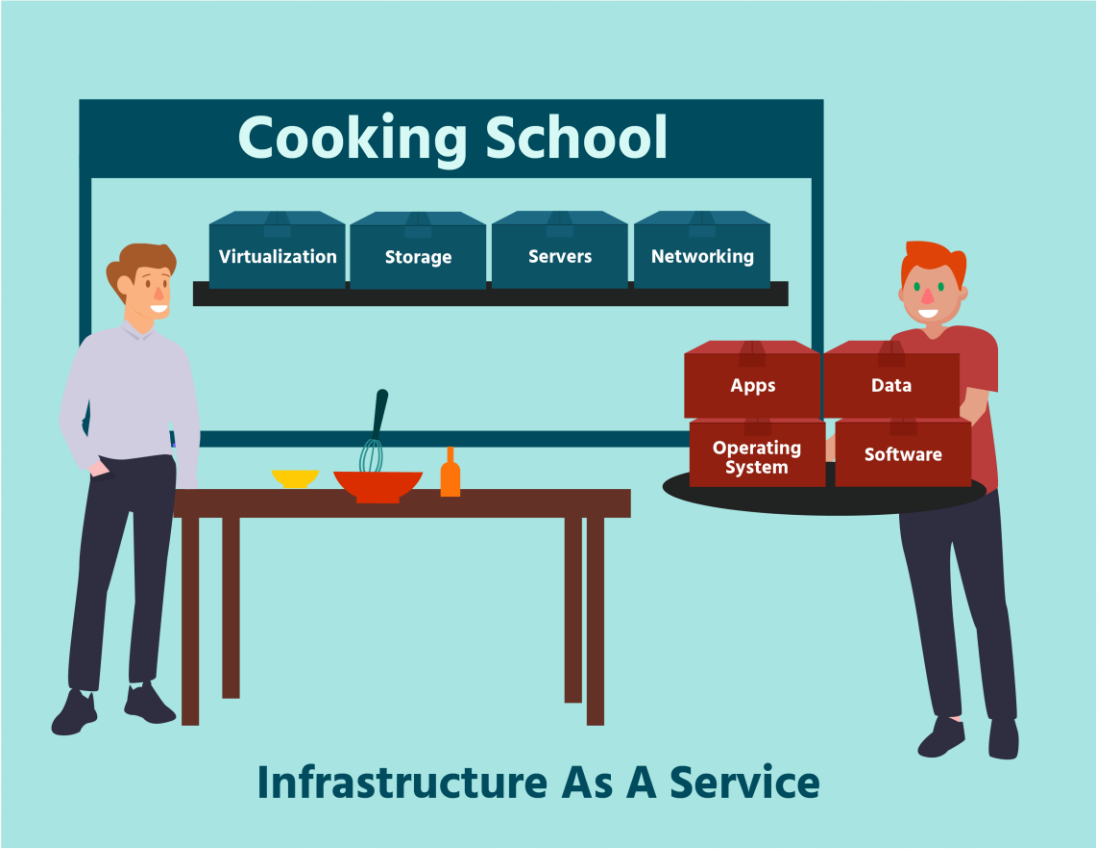
TYPES OF CLOUD COMPUTING

- **On-premise** (private cloud): You are responsible for **all** of your IT infrastructure (software and hardware)



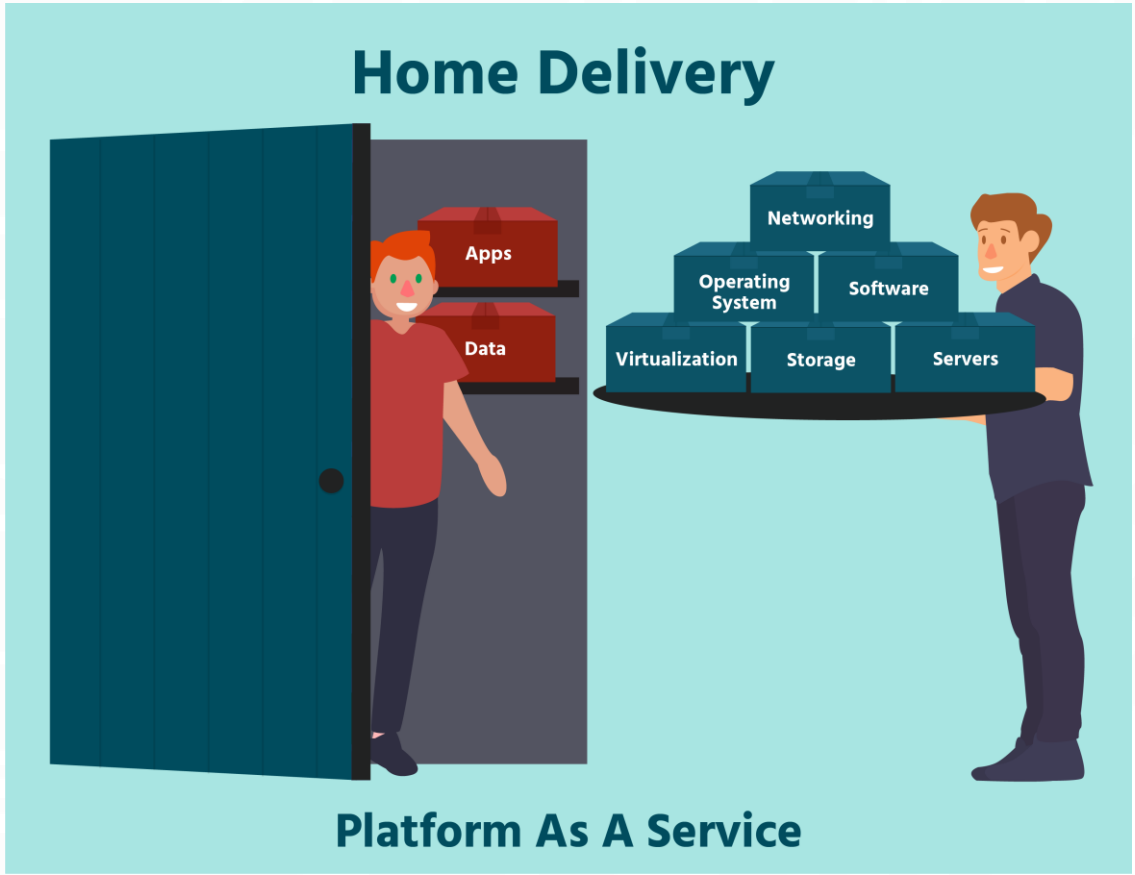
TYPES OF CLOUD COMPUTING

- **Infrastructure-as-a-service (IaaS)**: You are responsible for **applications**, **data**, **OS**, and **software**. Cloud provider manages your **virtualization**, **storage**, **servers**, and **networking**.



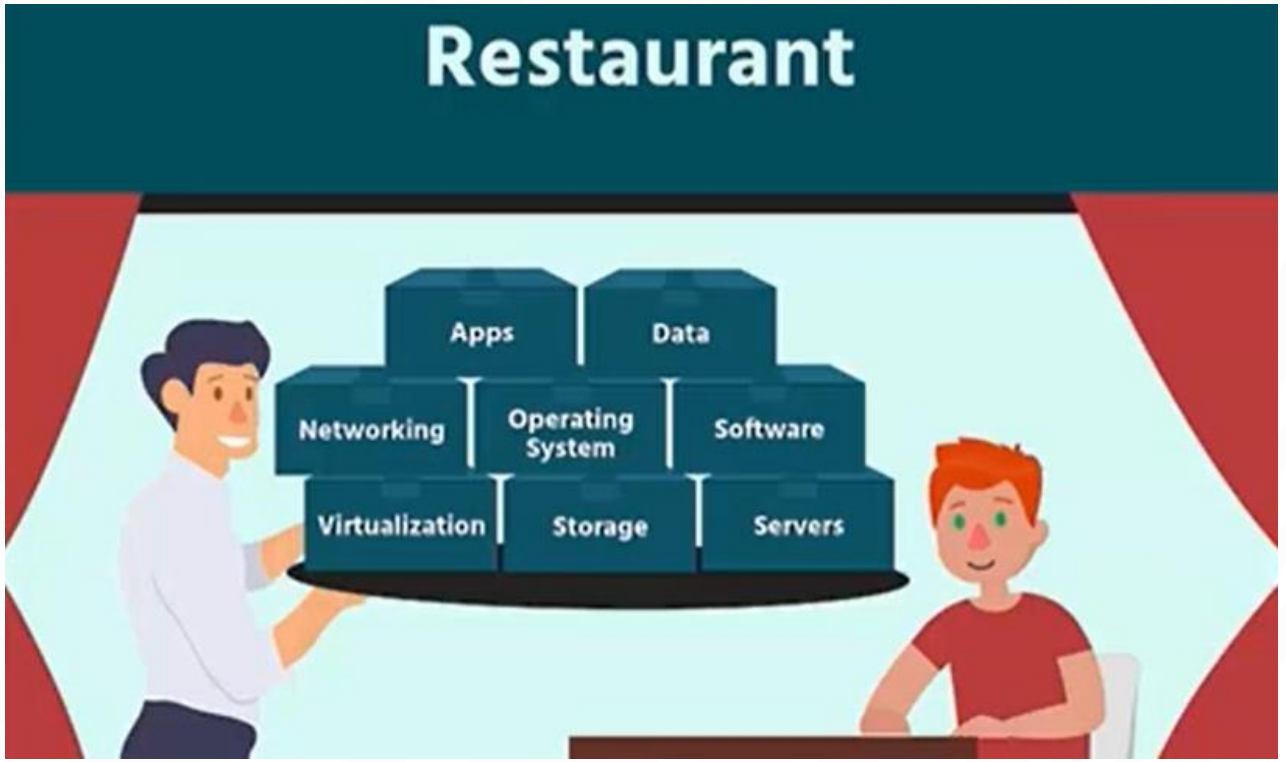
TYPES OF CLOUD COMPUTING

- **Platform-as-a-service (PaaS)**: You are responsible for **applications**, and **data**. Cloud provider manages your **OS**, **software**, **virtualization**, **storage**, **servers**, and **networking**.



TYPES OF CLOUD COMPUTING

- **Software-as-a-service (SaaS)**: You are not responsible for any infrastructure. Cloud provider manages your **OS**, **software**, **virtualization**, **storage**, **servers**, and **networking**.



CLOUD COMPUTING DEPLOYMENT MODELS:

- **Public.**
  - A cloud infrastructure that is open to use by the general public. It's owned, managed, and operated by a third party (or parties), and it exists on the cloud provider's premises.
- **Community.**
  - A cloud infrastructure that is used exclusively by a specific group of organizations.
- **Private.**
  - A cloud infrastructure that is used exclusively by a single organization. It may be owned, managed, and operated by the organization or a third party (or a combination of both), and it may exist on premises or off premises.
- **Hybrid.**
  - A cloud infrastructure that comprises two or more of the aforementioned deployment models, bound by standardized or proprietary technology that enables data and application portability (for example, fail over to a secondary data center for disaster recovery or content delivery networks across multiple clouds).

CLOUD DEPLOYMENT MODELS

- **Private Cloud**: organization has complete control over the IT infrastructure.

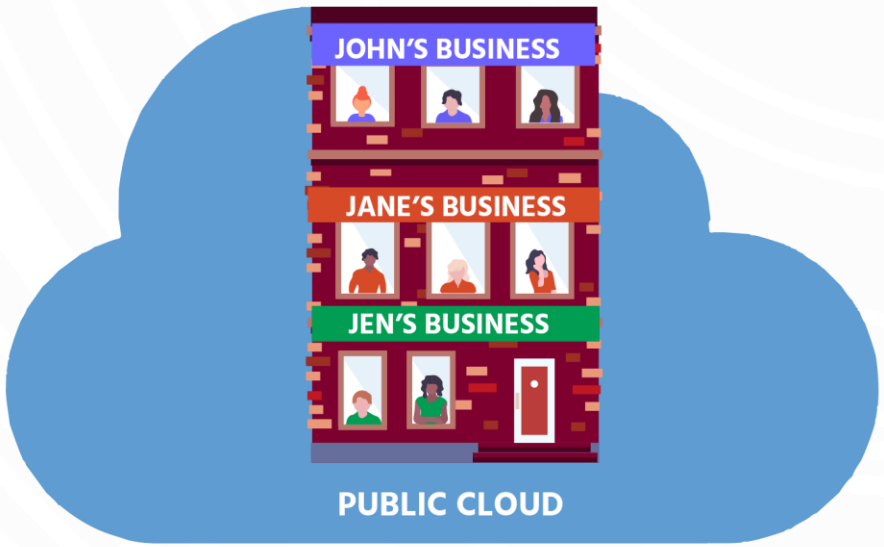


- **Community Cloud**: different organizations have control over the IT infrastructure.

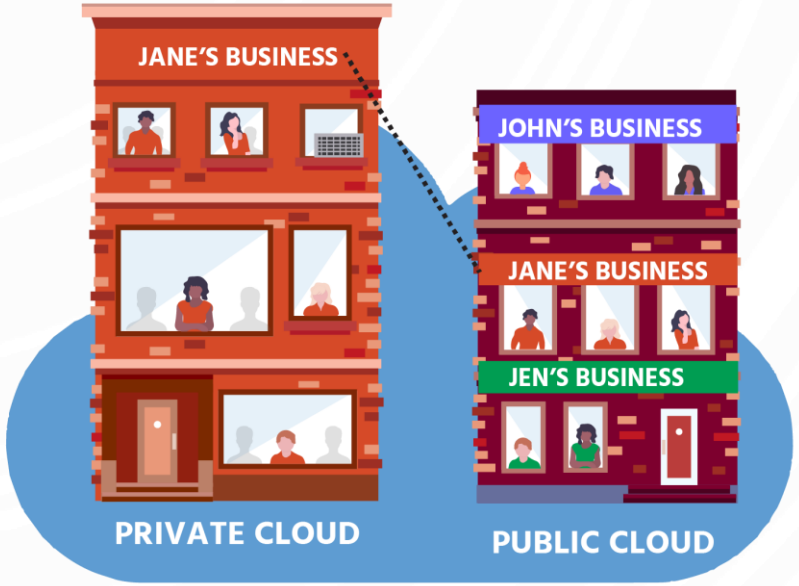


CLOUD DEPLOYMENT MODELS

- **Public Cloud**: the cloud provider owns the IT infrastructure and organizations rent IT resources in the cloud.



- **Hybrid Cloud**: organization has its own cloud but also rents IT resources on the public cloud.

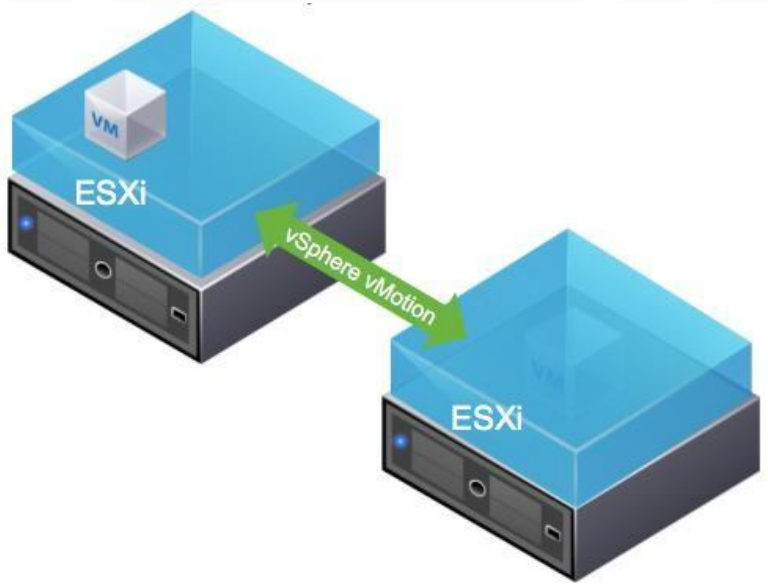


HYBRID CLOUD

VMWARE SOLUTIONS

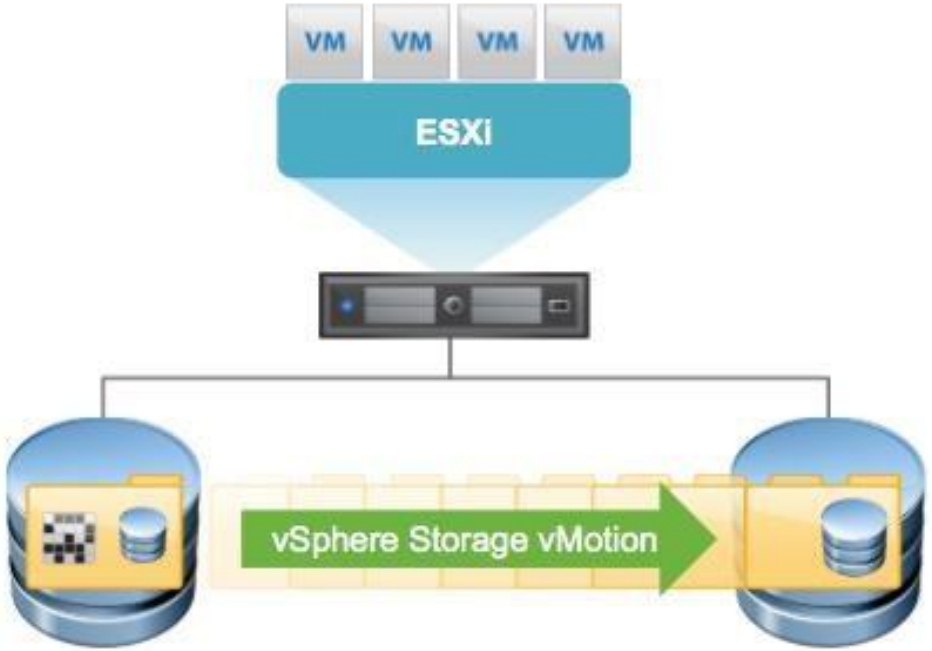
VMOTION

- Moves running virtual machines from one ESXi host to another ESXi host without service interruption (live migration)
- Increases **availability** of data and computing resources



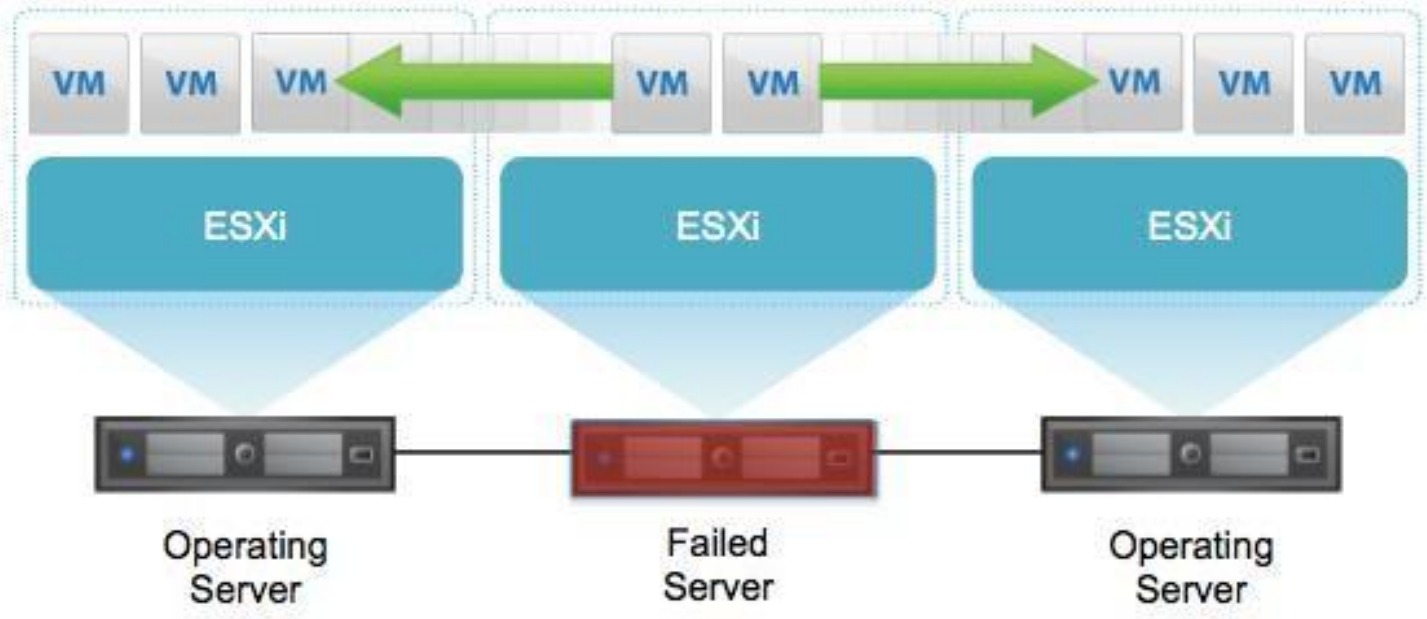
STORAGE VMOTION

- Moves the disks and configuration files of a running virtual machine from one datastore to another datastore without service interruption
- Increases **availability** of storage



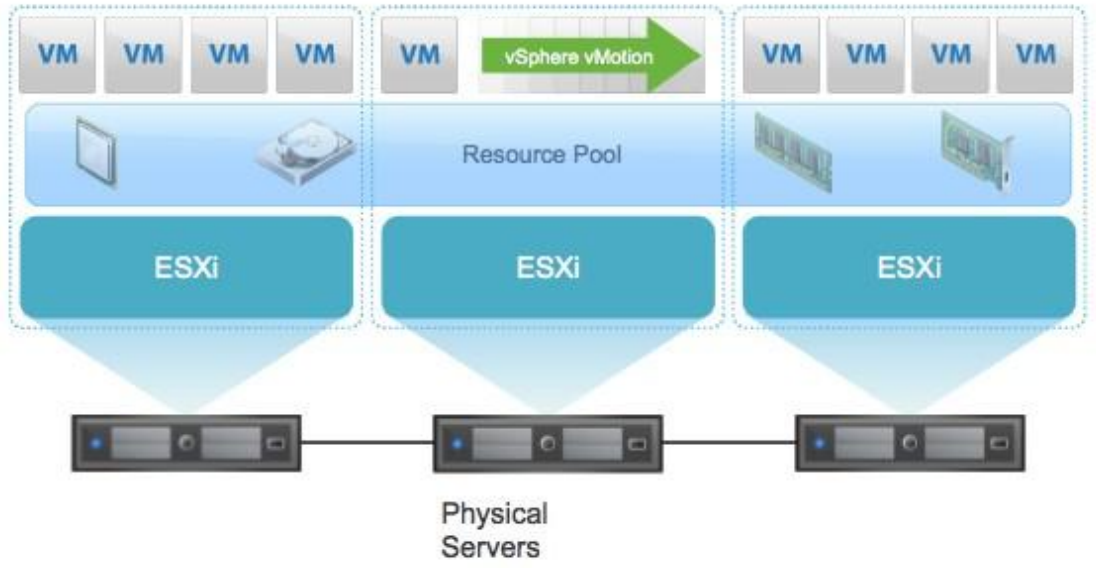
HIGH AVAILABILITY (HA)

- Pools servers (hosts) and the VMs that reside on them in a cluster so that in the event of a failure, the virtual machines on a failed host are restarted on alternate hosts



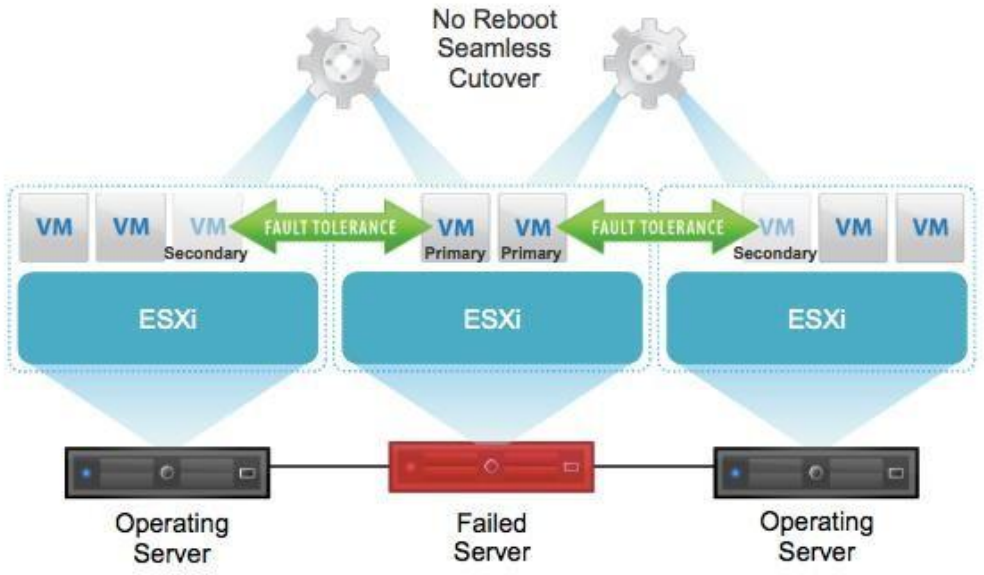
DISTRIBUTED RESOURCE SCHEDULER (DRS)

- What problem does DRS solve?
- HA clusters need to be monitored and managed. DRS implements a shared management interface so that the cluster's resources can be monitored and managed
- vSphere Storage Distributed Resource Scheduler provides the same solution for storage clusters



FAULT TOLERANCE (FT)

- A secondary copy of that virtual machine and its files is created on another ESXi host and datastore
- Using FT, the transfer to a different server is seamless and will not be noticeable to the end-user



REPLICATION

- vSphere Replication makes copies of VMs in a different physical location, useful for data protection and disaster recovery
- Works with vSphere Client to allow admins to monitor

